

Program overview

14-Dec-2022 12:20

Year 2022/2023
Organization Applied Sciences
Education Master Life Science and Technology

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master LST 2022							
General Compulsory Modules 2022							
LM3331	Modelling and Computation for Life Science & Technology	5					
LM3341	Microbial Omics	5					
LM3451	Bioprocess Integration	5					
LM3561	Ethical, Legal and Social Issues in Biotechnology	3					
LM3822	Design	12					
Electives 2022							
LM3311	Green Chemistry and Sustainable Technology	3					
LM3351	Topics in Biotechnology theory	0					
LM3356	Topics in Biotechnology practical	0					
LM3581NB	Metabolic Systems Biology	3					
LM3691	iGem	18					
LM3771	Protein Engineering	3					
Projects 2022							
LM3803	Company internship	15					
LM3804	Non-industrial Internship project	15					
LM3901	Master Thesisproject	45					
TUD4040	Joint Interdisciplinary Project	15					
Specialisation Modules 2022							
Cell Factory 2022							
LM3432	Analysis of Metabolic Networks	6					
LM3442	Metabolic Reprogramming	6					
LM3601	Molecular Biotechnology & Genomics	6					
Biocatalysis 2022							
LM3434	Advanced Protein Chemistry and Analysis	6					
LM3701	Advanced Enzymology	6					
LM3731	Advanced Biocatalysis	6					
Biochemical Engineering 2022							
LM3432	Analysis of Metabolic Networks	6					
LM3611	Microbial Community Engineering	6					
LM3741	Fermentation Technology & Environmental Biotechnology	6					
LM3751	Transport & Separation	6					
Scientific and Social Orientation 2022							
Study Abroad							
Management of Technology							
1st Semester MoT Modules							
MOT1412	Technology Dynamics	5					
MOT1421	Economic Foundations	5					
MOT1442	Social and Scientific Values	5					
MOT1461	Financial Management	5					
MOT1524	Leadership and Technology Management	5					
MOT2313	Research Methods	5					
2nd Semester MoT Modules							
MOT1003	Integration Moment	5					
MOT1435	Technology, Strategy and Entrepreneurship	5					
MOT1452	Inter- and intra-organisational decision making	5					
MOT1531	Digital Business Process Management	5					
MOT1534	High-tech Marketing	5					
MOT2421	Emerging and Breakthrough Technologies	5					
Honours Programme 2022							
Collective Part 2022							
UD2010	Critical Reflection on Technology	5					
Individual Part 2022							
1. Company Oriented HPM 2022							
AS1011HPM	Applied Sciences Company Project	12					
AS1021HPM	Applied Sciences Honours Classes	3					
2. Consistent package of challenging courses 2022							
3. Highlights in Biotechnology 2022							
LM3351	Topics in Biotechnology theory	0					

LM3356

Topics in Biotechnology practical

0



4. Research Oriented HPM 2022

AS1031HPM

Applied Sciences Research Project (9-15 EC)

9



Year	2022/2023
Organization	Applied Sciences
Education	Master Life Science and Technology

Master LST 2022

Contact for students	info-lst@tudelft.nl
Program Title	MSc programme Life Science & Technology
Director of Education	Prof. dr Ulf Hanefeld, U.Hanefeld@TUDelft.nl
Program Coordinator	Andrea Witkam, a.a.witkam@tudelft.nl
EC Program	120 ECTS, Language of instruction: English
Program Start (Study Year)	30 August 2021
Prerequisites	Below you will find the description of the MSc program Life Science & Technology 2021 - 2022. Entry levels are described on the TU Delft website: TU Delft-> Education -> Programmes -> Master programmes-> Life Science & Technology
Program Goals	The programme aims: - To instil scientific knowledge, insight, as well as methodical, technological and communicational skills that correspond to the specialised field of the programme; - To gain an academic attitude with a critical, scientific and creative way of thinking, based on awareness of the ethical and social aspects relating to ones own research, and insight into the professional situation and into the consequences of ones actions on others; - To prepare for a scientific career and further education, especially for a doctoral education (PhD); - To prepare for a career in society. The graduate will be able to pursue an independent profession.
Program Structure 1	The curriculum covers two years (120 ECTS) YEAR 1: 1. General Compulsory Modules (15) 2. Compulsory Specialization-specific Modules (18) 3. Electives (12) Electives can be chosen from courses of other specializations and/or from a list of recommended courses and/or from any Applied Science Master course and/or (with permission from the board of examiners) from other Master's courses. The list of preferred electives will be published on Brightspace -> Master Life Science & Technology -> Content -> Study Info. 4. LM3822 Design Project (12) + LM3551 Ethical, Legal and Social Issues in Biotechnology (3) YEAR 2: 5. LM 3901 Master Research Project (45) The research project is specialization-specific 6. LM3803 Company Internship (15), or TUD4040 Joint Interdisciplinary Project (15) LM3804 non-industrial internship* *In exceptional cases, the programme coordinator may allow a student to carry out a non-industrial internship project (LM3804, 15 EC) with learning outcomes equivalent to those of the Company Internship LM3803.
Specialisation	The MSc programme Life Science & Technology has 3 specializations: BIOCATALYSIS - coordinator Frank Hollmann, F.Hollmann@tudelft.nl (Appointments: send an email with your question(s). Put 'meeting request' in the subject of your mail) CELL FACTORY - coordinator Pascale Daran - Lapujade, P.A.S.Daran-Lapujade@tudelft.nl (appointments through M.C.VanderBurg@tudelft.nl) BIOCHEMICAL ENGINEERING - coordinator Walter van Gulik, W.M.vanGulik@tudelft.nl (Appointments: send an email with your question(s). Put 'meeting request' in the subject of your mail)
Options	Scientific & Social Orientation Students are allowed to substitute 30 EC of the above-mentioned MSc programme LST by the following Special Programmes. - Track 'Science Education' (in Dutch) (info-sec@tudelft.nl) - Track 'Communication Design for Innovation' (info-sec@tudelft.nl) - Study abroad (InternationalOffice-TNW@tudelft.nl). - One semester of the MSc programme Management of Technology* (Andrea Witkam, a.a.witkam@tudelft.nl) ATTENTION!!! Students with the exam programme of 2021 substitute the elective part (12 EC), the company internship (15 EC), and 3EC of their MSc End Project (MEP) = 30 EC. Students with an exam programme of 2020 or older: they substitute the elective part (12 EC) and the company internship (18 EC) = 30 EC. These special programmes must be approved by the coordinator of the specialization. *Limited # of places available. Application deadlines: 15th November (for semester 2) and 1st of May (for semester 1)

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General Compulsory Modules 2022

LM3331	Modelling and Computation for Life Science & Technology	5
Responsible Instructor	Dr.ir. C. Haringa	
Responsible Instructor	R. Gonzalez Cabaleiro	
Contact Hours / Week x/x/x/x	4-6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	BSc degree (Mathematics 1st and 2nd year; Programming or Algorithmics; Transport phenomena; Biotechnology; Chemistry)	
Course Contents	The course deals with deriving models from biosystems (process, cell, enzymes), mathematical formulation and numerical solutions. Model formulation: Mass balancing for biosystems Mass balancing in space and time Mechanistic modeling for kinetics Thermodynamics Mathematical solutions: Linear equations Non-linear equations ODE systems PDE systems Optimization, parameter estimation Sensitivity analysis Model validation and error propagation	
Study Goals	After this course the students should: - be able to formulate coherent mathematical models - be able to understand the mathematical formulation of relevant algorithms - be able to identify the type of computational problem - be aware of the computational tools existing for the solution - be able to solve different model systems by creating computer programs (code) in Python - be able to critically analyse the results obtained and formulate a report, conclusions and recommendations. - be able to apply basic "best practices" for code development and sharing, and data treatment	
Education Method	The course consists of a combination of the following methods: - lectures (theory) - computer labs (practice) - computer assignments (application)	
Assessment	The course will be evaluated via: - Individual assignments (total: 25%) - Group assignment including presentation (25%) - Final exam (50%)	

LM3341	Microbial Omics	5
Responsible Instructor	Dr. M. Pabst	
Responsible Instructor	Prof.dr.ir. J.G. Daran	
Module Manager	Dr. M. Pabst	
Module Manager	Prof.dr.ir. J.G. Daran	
Contact Hours / Week x/x/x/x	4-6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	BSc degree (Introduction to Algorithmics and Programming; Statistics; Molecular Analytical Methods; Bioinformatics course; Cell Biology; Microbial Physiology)	
Course Contents	Omics perform the collective characterization and quantification of biological molecules that translate into the structure, function, and dynamics of organisms. This course aims to provide fundamentals and practical aspects of the state-of-the-art of omics technologies used to study pure and mixed microbial cultures such as whole (meta)genome sequencing, (meta)proteomics and metabolomics. The course also provides examples of how those data can be used to obtain information on the structure and function of microorganisms. Fundamentals of omics techniques: Whole-genome and shotgun metagenome sequencing; Large-scale (meta)proteomics; (Meta)metabolomics; Brief introduction into other, more specialized omics approaches. Omics data handling and interpretation: Introduction to generic omics processing pipelines (where applicable); Data handling such as interpretation of omics data (e.g. discussing published datasets).	
Study Goals	After this course, the students will be able to: - evaluate the advantages and limitations of various omics techniques; - interpret data obtained from various omics techniques; - design appropriate omics experiments to study pure and mixed microbial cultures; - work in small groups and communicate large scale omics data.	
Education Method	Lectures, tutorials and exercises.	
Assessment	Assignments (20%) and final written exam (80%). (retake will be on written exam only)	

LM3451	Bioprocess Integration	5
Responsible Instructor	Dr.ir. A.J.J. Straathof	
Module Manager	Dr.ir. A.J.J. Straathof	
Contact Hours / Week x/x/x/x	4-6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This is an introductory, mandatory course for the MSc LST and prepares especially for the Design project, for Fermentation Technology & Environmental Biotechnology, and for Transport & Separation	
Expected prior knowledge	Basic biosciences, process technology, bioprocess design, chemistry and mathematics as required for admittance to the MSc programme Life Science & Technology. The expected biochemical engineering background is: P.M. Doran, Biochemical Engineering Principles, Academic Press 1995, chapters 1-9 and 11, except pages 139-163. Or the equivalent in the 2nd edition of this book. The expected bioprocess design background involve capability - to study and use industrial biotechnology literature - to calculate fermentation stoichiometry - to formulate simple bioprocess block schemes - to calculate mass flows in such processes and assess associated economic/ecologic/social impact	
Course Contents	- Short recap of bioprocesses in industrial and environmental biotechnology - Stages in the bioprocess design cycle that are treated in this course: design definition, synthesis, analysis, process assessment - Flow sheeting and process simulation using dedicated software (SuperPro Designer), to solve mass and energy balances of bioprocesses, to size equipment, and to perform economic assessment - Operation modes: batch/continuous; cocurrent/countercurrent/crosscurrent - Bioprocess unit operations and their short-cut design: enzymatic conversion, fermentation, sedimentation, centrifugation, filtration, membrane permeation, L/L extraction, adsorption, ion-exchange, chromatography, evaporation, distillation, precipitation, crystallization - Selecting and sequencing unit operations; integration of bioprocess functions; in-process recycles; waste destinations - Batch scheduling and debottlenecking - Energy streams in bioprocesses; heat integration - Economic assessment method of SuperPro Designer; sensitivity of process performance to microbial and enzymatic parameters and process settings	
Study Goals	After the course the student should be able to - describe the functions and the functioning of the different parts of biotechnology processes - draw block schemes of bioprocesses suitable for detailed analysis - use software for bioprocess simulation in order to obtain mass and energy balances, equipment sizing, batch scheduling, heat integration, and economic assessment - perform off-line short-cut calculations on sizing the most relevant unit operations in biotechnology, on batch scheduling, on heat integration, and on economic assessment - identify semi-quantitatively the potential impact of protein or metabolic engineering on overall improvements of a production process	
Education Method	- Lectures - Tutorial - Training exercises in classroom - Assignments	
Computer Use	SuperPro Designer (will be instructed)	
Literature and Study Materials	- Lecture slides handouts - Supporting documents - Training exercises + solutions - Old exams + solutions All will be posted on Brightspace during the course.	
Assessment	During the course three assignments on a case study in teams of up to three students (unrounded assignment grade counting for 50% of the unrounded overall grade) and a final individual written exam (unrounded exam grade also counting for 50%). But if the rounded overall grade is >5.5 while the rounded grade for the exam is <5, the rounded overall grade will be decreased to 5.5. Calculation examples: - Assignments 4.70, exam 9.30, rounded overall grade 7.0 - Assignments 9.30, exam 4.70, rounded overall grade 5.5 The exam has a resit. Then the assignments grade will be kept for recalculating the overall grade. Assignments can be redone only in a next year. Then, the exam grade will be kept, unless the exam is also redone.	
Enrolment / Application	Before the first lecture via Brightspace	
Studyload/Week	4-6 contact hours per week; furthermore, about 8 hours/week for self study and assignments. At the end, a full week for the exam incl. preparation	
Continuing Courses	Design project LM3822	
Schedule	Contact hours that involve lecturing and exercising: Week 1.1: Introduction (1h), Bioprocess simulation training (5h) Week 1.2: Biokinetics and bioreactors (4h) Week 1.3: Batch scheduling (2h); Solid/liquid separation (2h) Deadline assignment 1 Week 1.4: Contacting and extraction (4h) Week 1.5: Evaporation/drying/distillation (2h); Adsorption/ion-exchange/chromatography (2h) Deadline assignment 2 Week 1.6: Precipitation/crystallization; Membrane operations; Design definition; Process assessment (each 1h) Week 1.7: Bioprocess energy (2h); Bioprocess synthesis (2h) Deadline assignment 3 Some swapping may occur between weeks 1.5 and 1.6 depending on the needs of assignment 2. Weeks 1.3, 1.5, and 1.7 include also 2h support for assignments.	

A detailed lecture schedule will be published on Brightspace before the course starts.

LM3561	Ethical, Legal and Social Issues in Biotechnology	3
Responsible Instructor	Drs. L. Asveld	
Instructor	Drs. L. Asveld	
Contact Hours / Week x/x/x/x	0/0/0/8 hrs/week	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	This course is open to MSc students with a relevant background (LST, Chemical Engineering, nanobio).	
Course Contents	In this course you will explore ethical, legal and societal issues of biotechnology that will be relevant to your future work as a professional in biotechnology. The course will introduce concepts in ethics in the life sciences and familiarize you with some of the most prominent ethical theories and their applications. The course will also stimulate you to critically reflect on the design assignment you do parallel to this course and to integrate a wide range of societal considerations into your design assignment. This course is a combination of lectures and seminar sessions, coupled with a reflection on your design project and other topics of interest.	
Study Goals	After having completed the course you: - will be able to recognise and analyse ethical, legal and societal issues related to biotechnology. - will be able to apply methods to create societally robust and ethically acceptable designs	
Education Method	The course is highly interactive, combining lecture and seminar discussions with student presentations.	
Literature and Study Materials	Selected texts and other online material available on Brightspace	
Assessment	50 % group assignment related to your design project (1 presentation) and 50 % individual written assignments (1 essay).	
Enrolment / Application	Enrolment via Brightspace (courses > Master > LS&T) is required one week before the first lecture.	
Schedule	Q4	
Location	Delft	

LM3822	Design	12
Responsible Instructor	Prof.dr.ir. L.A.M. van der Wielen	
Instructor	Dr. A.A. Kiss	
Instructor	Dr. O. Thum	
Contact Hours / Week x/x/x/x	0/0/0/32	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Parts	The first activity in the Design Project is to translate the concise description of the industrial assignment to a result that is agreed upon with the principal. For this, you need to familiarize yourselves with the topic and you will probably find that multiple options exist to solve the specific problem. You have to reduce the number of feasible options to a workable number, and therefore you have to make choices. The basis for the reduction and selection of options has to be as quantitative as possible. The combined information forms the so-called Basis of Design. This is a (short) intermediate report that describes the specifications of the design, the selection/reduction process, and its outcome. The purpose of the Basis of Design is to agree upon the envisaged results of the project, first among the team members, and then, most importantly, with the principal. Acceptance of the Basis of Design is required to continue the project. The remainder of the project period is used to work out the selected options, which can follow different lines. You will find that, in order to be efficient, tasks have to be divided among team members. Still, all of you will be responsible for the result: every team member must be able to defend the complete work, not only his or her share. The end result is presented in the form of a written Final Report, which must be presented and defended before the coaches and the industrial principal. You can see them as the Board of Directors of your company, who have to decide if they want to take the financial risk of investing in your ideas.	
Course Contents	A characteristic element of an engineering education is training in engineering design. This implies integration of existing knowledge, exploration of new fields, making choices and relating the result to a range of constraints (economy, environment, society, safety, validation, quality assurance etc.). For the MSc Life Science & Technology, the design project can comprise a wide range of subjects: specific bioproducts (e.g. a pharmaceutical or food additive), a specific bioprocess (for fine or bulk chemicals, environmental), a (metabolic route in) micro-organism or (diagnostic) equipment. Depending on the topic, a team of students from different LST-profiles collaborate in a structured manner towards a complete design to be presented in a written report as well as in an oral presentation. The team has to organize itself such that the eight weeks are used in the most efficient manner. The team presents the project progress in weekly meetings with the coaches.	
Study Goals	After the course the student should be able to - Integrate and apply theoretical knowledge in practical projects - Creatively generate options and systematically make choices based on qualitative and quantitative arguments, working with conflicting requirements - Implement Project Management and Team Work	
Education Method	Project Team work, with additional: Lectures, Assignment from industry, weekly meetings with coaches, regular meetings with industrial principal	
Literature and Study Materials	LST Design Project Manual (and presentation), Project Management Manual (and presentation)	
Assessment	1. Theory: this includes the quality of the content - 70% 2. Practice: this includes project management, interaction with the principal and the group process -20% 3. Reporting: this includes the Basis of Design Report, the Final Report and the presentations 10%	
Enrolment / Application	As the coaches need to know the number of students in order to arrange the projects with the industries, you have to enroll on Brightspace roughly a month before start (March 15th in 2022).	
Remarks	On the planning: The Design Project is credited 12 EC, so 8 weeks of full time work, that are distributed over a period of 10 weeks since the period includes Ethics (2 weeks). This means you only have two to three weeks to sort out the information for your Basis of Design; you will have four to five weeks for working out the selected options, and you will need the remaining one or two weeks for reporting (and probably correcting some calculations). Remember the result has to be quantitative, or at least semiquantitative. Furthermore, allow sufficient time for writing the reports. You have to make your own planning, together with the members of your team and in accordance with the rest of your activities. We expect you to have none or only one other course in parallel with this project, and advice you to work on this as much as possible in a block form. A tentative planning looks as follows: Weekly Progress meeting with coaches Week 1 or 2 Kick-off meeting (with principal and coaches) Week 3 or 4 Base of Design presentation (with principal and coaches) Week 8 (10) Finalization of project Week 8 (10) Final Presentation (with principal and coaches)	
Schedule	8 weeks in the 10 weeks of the 4th period	

Electives 2022

Introduction 1

A detailed list with suggested electives is published on the Brightspace Organisation Master Life Science & Technology (study information) or at request (info-lst@tudelft.nl).

IMPORTANT: all (elective) courses have to be approved by the coordinator of your specialization. These electives should be mentioned in the MEP Application Form (see also Brightspace organization Master LST --> Content --> Study Info --> Forms)

- 1) The elective course should have an academic level and have ECTS associated with it.
- 2) The student should be able to convince the coordinator of the specialization, that the course adds to his/her prior knowledge and that the course contributes to the development of knowledge or skills that are relevant for his/her future
- 3) Courses at other universities in the Netherlands and abroad (academic courses abroad are sometimes combined with a stay abroad for an industrial internship) should always be discussed with your specialization coordinator.

LM3311	Green Chemistry and Sustainable Technology	3
Responsible Instructor	Dr. F. Hollmann	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
	5	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Organic Chemistry (LB1045 or 4051OCSTRY) Chemical Biology (LB2621) OR Catalysis (4052KATALY)	
Course Contents	Depleting fossil fuels and environmental concerns force us to change the way we produce our goods. The transition of the chemical industry to becoming sustainable represents one of the major challenges to be faced in the 21st century. Catalytic reactions will increasingly substitute the existing energy- and resource-intensive production procedures; hence reducing or preventing toxic pollutants. Biotechnology will play a major role in this transition and complement advanced chemical production technologies. This course will highlight the emerging role of catalysis in the chemical industry. A major focus will lie on biotechnological production of chemicals (fermentation processes, isolated enzymes). These methods will be related to their chemocatalytic counterparts. Thus, ample examples including the underlying mechanistic principles - of bio- and chemocatalytic processes will be discussed. Finally, the emerging field of renewable feedstocks will be discussed not only from a chemical standpoint, but also on socioeconomical and ecological grounds. The 12 Principles of Green Chemistry will serve as guideline to assess the scope and limitations of the examples discussed.	
Study Goals	After attending this course the student will: (1) have detailed knowledge of sustainable catalysts (enzyme- and chemo-catalysis) (2) know industrial examples of sustainable manufacturing practice (3) be able to use the 12 Principles of Green Chemistry to assess the sustainability of a given process (4) be able to critically assess the purport of green claims	
Education Method	Lecture Sheets, specialized books, articles and patents	
Literature and Study Materials	selected articles will be available via Brightspace	
Books	suggested reading: 'Green Chemistry - An Introductory Text' (2nd ed., M. Lancaster) RSC Publishing (ISBN: 978-1-84755-873-2) and/or 'Green Chemistry and Catalysis' (Green Chemistry and Catalysis R. A. Sheldon, Isabella Arends, Ulf Hanefeld), Wiley VCH (SBN: 978-3-527-30715-9)	
Assessment	(1) Written exam (digital off-campus (open book) or on-campus, depending on the COVID-19 measures), 75% of the overall grade (2a) only for SET students: written assignment about a topic of Green Chemistry (groups of 3-4 students), 25% of the overall grade (2b) for non-SET students: group presentation about a topic of Green Chemistry (groups of 4-5 students), 25% of the overall grade	

LM3351	Topics in Biotechnology theory	0
Responsible Instructor	C.E. Paul	
Responsible Instructor	Prof.dr. U. Hanefeld	
Contact Hours / Week x/x/x/x	0/0/x/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	BSc degree LST or similar. The course focusses on students joining in February rather than at the general starting time in September, at exchange students and at honours students.	
Course Contents	THE COURSE LOAD VARIES BETWEEN 3 - 5 EC The course will focus on a current high profile topic in the life sciences & technology. This topic will alter every second year and will be chosen by the teacher of the course. Details will be given by the teachers. The topic will be taught in lectures and self-study and will be finally evaluated via an essay of adaptive length to suit the desired number of ECTS. 2022/2023: Biocatalytic retrosynthesis. The course will focus on the retrosynthetic approach using enzymes. Recent development of biocatalysts that can be applied in organic synthesis allows to rethink the way in which target molecules might be constructed. Principles of biocatalytic retrosynthesis with consideration for strategic disconnections, functional group interconversions, and structure-guided analyses will be studied. A work plan will be made on how to disconnect a target molecule into identifiable building blocks that can be produced by biocatalysts.	
Study Goals	Explore an unknown current research topic via literature, guided reading and discussion After the course the student will have developed in depth knowledge on the topic of the course and have mastered the skills of writing an essay. This includes: - Understanding why the topic is considered high profile - Precise and concise formulation of the topic and its relevance to Life Science & Technology and to society in general - Explore an unknown current research topic via literature, guided reading and discussion	
Education Method	Lectures, Self-study, research plan	
Literature and Study Materials	References made available on brightspace	
Assessment	Essay of adaptive length to suit the desired number of ECTS (THE COURSE LOAD VARIES BETWEEN 3 - 5 EC)	
Special Information	The course load varies between 3 - 5 EC, depending on the 'Topic of the year' and/or the desired length of the essay.	

LM3356	Topics in Biotechnology practical	0
Responsible Instructor	C.E. Paul	
Responsible Instructor	Prof.dr. U. Hanefeld	
Contact Hours / Week x/x/x/x	0/0/x/x	
Education Period	3 4	
Start Education	3 4	
Exam Period	Exam by appointment	
Course Language	English	
Required for	The course focusses on students joining in February rather than at the general starting time in September, at exchange students and at honours students.	
Expected prior knowledge	BSc degree LST or similar. Topics in Life Science & Technology theory.	
Course Contents	THE COURSE LOAD VARIES BETWEEN 10 - 12 EC, depending on the 'Topic of the year' and/or the desired length of the essay. The course will focus on a current high profile topic in the life sciences & technology. This topic will alter every second year and will be chosen by the teacher of the course. Details will be given by the teachers. The topic will be taught in laboratory work and self-study and will be finally evaluated via the BEP rubric to suit the desired number of ECTS. This year 2022/2023 the biocatalytic retrosynthesis approach is the target of the course. In the theory part of the course the students will have developed a plan on how to disconnect a target molecule into identifiable building blocks that can be produced by biocatalysts. In the second part this theoretical exercise will be put into practice.	
Study Goals	After the course the student will have developed and applied in depth knowledge on the topic of the course and have performed an independent piece of practical research. This includes: - Understanding why the topic is considered high profile - Precise and concise formulation of the topic and its relevance to Life Science & Technology and to society in general - Explore an unknown current research topic via literature, guided reading and discussion - Formulating a research goal based on the above - Formulate a research plan on how to perform this research and execute it - Write a report on the achieved results	
Education Method	Practical work in the laboratory	
Literature and Study Materials	Made available on brightspace	
Assessment	Evaluation with rubric as applied to the BEP (BSc end project)	
Special Information	The course load varies between 10 - 12 EC, depending on the 'Topic of the year' and/or the desired length of the essay.	

LM3581NB	Metabolic Systems Biology	3
Responsible Instructor	Dr. G.E. Bokinsky	
Course Coordinator	Dr. G.E. Bokinsky	
Contact Hours / Week x/x/x/x	0/0/2-3/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Systems biology is an integrated approach of experimental and mathematical methods that aims a comprehensive understanding of how biological functions arise and how they are regulated. We will focus on different levels of cellular regulation in bacteria: metabolism, gene expression, and global physiology, while emphasizing interconnections between the regulatory levels to illustrate how stable cellular phenotypes occur, and how these interconnections give rise to seemingly-miraculous properties, like growth rate maximization. The course will consist of a mixture of lectures and student presentations and discussions of current literature. Experimental and modelling approaches will be discussed. For unravelling how cell functions are regulated, different platforms have been developed, among which are techniques named - omics technologies, e.g. metabolomics, transcriptomics, genomics and proteomics. However, data alone are insufficient for understanding how a metabolic pathway or a signalling cascade gives rise to different phenotypes. Using literature presentations and discussion, this course introduces the uses measurement technologies for the purpose of: - Quantifying metabolites within bacteria - Quantifying genetic regulation in bacteria - Quantifying single-cell gene expression - Identification of interactions between genes and metabolism The course presents: - Measurement techniques and data interpretation for protein and transcript levels at both ensemble and single-cell level - Principles and functions of gene expression networks, emphasizing autoregulation and regulatory dynamics - Using modelling at different levels of detail to explain cellular behaviors and adaptive advantages - Using modelling to identify sites of regulation in metabolic pathways Note that online or in-person classroom attendance is mandatory at all sessions.	
Study Goals	After this course the students should be able to: - Understand how models published in system biology journals are used to explain experimental results - Predict dynamics of cellular networks in response to perturbations - Explain how regulatory connections and dynamics result in a defined phenotype - Explain how data and models are used to investigate cellular behavior - Critically evaluate literature papers in field of systems biology - Communicate findings from literature (and criticism of the literature) to an audience of non-experts	
Education Method	Lectures Peer lectures Computer practicals Classroom Discussions	
Assessment	- Computer Exercise on gene transcription (25 %) - Literature case study: Oral presentation of one journal article and leading a classroom discussion, either alone or as part of a group (50%) - Participation in classroom discussion (15%) - Short written analyses of assigned reading (1 paper per week) (10%) There will be no final exam.	

LM3691	iGem	18
Responsible Instructor	Dr. C.J.A. Danelon	
Instructor	Dr. C.J.A. Danelon	
Practical Coordinator	Ing. E. Yildirim	
Assistant	M. Holub	
Assistant	E. Godino	
Assistant	A.M. Restrepo Sierra	
Assistant	C.C. Koster	
Assistant	Dr.ir. B.F.H.J. Bouchaut	
Gast Instructor	Dr.ir. S.J.J. Brouns	
Gast Instructor	T. Idema	
Contact Hours / Week x/x/x/x	n.a.	
Education Period	Different, to be announced	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	iGEM, the international genetically engineered machines competition was started in 2003 at MIT, Boston. The objective of the competition is to design and build an engineered biological system using DNA technologies over the course of the summer. The engineered systems (mostly E.coli) will be constructed from standard biological parts (plasmids), called BioBricks in the iGEM community. The team chooses their own project: It can be a very fundamental project, such as biological information processing, or a more applied project, with uses for medical, environmental or manufacturing purposes. During the course of the summer you will design your experiments, construct parts (plasmids) and validate the constructs using (new) characterization assays. Besides the molecular work an important aspect of iGEM is modeling. Comparable to engineering approaches, the parts' function and interaction are modeled to predict their behavior and foster the understanding of relevant characteristics. Thereby the modeling assists the design and construction of improved parts and experimental methods. Alongside the scientific work, you will be cooperating with scientific and non-scientific communities to promote and communicate your project. The project will be publicized through public relations efforts, social media, and the teams own wiki page. Funding for the team will be raised by the students from university groups and local industries. The project will also have a policy and practice component, in which the social, legal, and ethical issues of the project are considered, through considerations of the possible risks of the project, interviews with relevant end-users and affected community members, and designing of appropriate safety and deployment mechanisms. The team will be advised by several experienced faculty members, technicians, and graduate students. A diverse group of students is optimal, with backgrounds ranging from biology and bioinformatics to chemistry, physics, and engineering disciplines. You will have the opportunity to travel to Boston (or Paris) and interact with and compete against >250 universities from all around the world.	
Study Goals	After this course the students are able to: - Develop a relevant project idea. - Plan and perform experimental work, esp. cloning (digestion, ligation, transformation). - Develop assays for the characterization of gene functions (e.g. reporter genes, enzyme activities, growth phenotypes, etc.). - Evaluate experimental observations using mathematical modeling - Test hypothesis using experimental design. - Document your project and progress (web, wiki, poster, presentations) - Communicate the project and results to the scientific and non-scientific audience. - Organize the work in a team. - Raise funding from governmental organizations and industry.	
Education Method	Participation in the TU Delft iGEM team	
Assessment	The performance is monitored during the project course. Depending on the major contribution, the following aspects will be assessed: Intellectual skills - Logic, reflection - Independent formation of opinion - Motivation/interest/enthusiasm - Application of theory from scientific literature Scientific method and approach - Independent approach to work - Systematic approach - Takes initiative - Works well with others and participates in group work - Critical attitude (in a positive sense) Competence in research and/or design - Practical skills - Analytical skills - Structured, exact, and efficient working style - Creativity Reporting - Able to document observations & results (wiki) - Logical structure - Consistency - Accurate representation of the project's progress (group meetings, wiki, small paper reports) Presentation, communication and discussion - Level of (final) presentation - Handling of critical and scientific questions - Depth and argumentation in answering questions	
Enrolment / Application	This MSc project is open for students from all MSc programs at TU Delft.	

	To apply as member of the iGEM team, please write an application letter including: - motivation to participate in an iGEM competition - describe your favorite project idea - which role in the team would suit you most? It is MANDATORY to contact your academic counsellor prior to applying. Interviews prior to the formation of the team are part of the application procedure. In this interview supervisors will discuss with you your motivation, academic level, but also your 'soft' skills and other personal capacities, such as communication & writing skills, time-management, pro-activeness, being a team player, etc. Please also include a C.V. with a list of courses in the application. It is recommended that you contact your study-advisor prior to applying. NB-students, planning the iGEM project as their minor, should certainly contact the study-advisor or program coordinator in advance.
Remarks	The iGEM team is limited to 12 students. Students are welcome from all study programs of TU Delft. For LST master students: The course grants 18 ECTS. 12 ECTS can be included in the LST master study program (by application to the board of examiners, e.g. to replace the design project). The remaining 6 EC are extracurricular. This means your exam program will be set on 126 credits in total. Excellent students are encouraged to apply for the TU Delft honours program. Please contact the director of education (Ulf Hanefeld) IN ADVANCE. The iGEM project can be performed as part of an individual honours program, under the condition that the Design project will be carried out as well! For Applied Physics master students: The course grants 18 ECTS. Upto 12 ECTS may be included in the Applied Physics master study program as a General Elective. The remaining 6 ECTS are extracurricular. Alternatively, the course can be done in an individual honours program by students qualifying for the TU Delft honours program. For bachelor students: the course grants 18 ECTS. These can be included in the TU Delft minor program DREAM-team (WB-Mi-121-12). They require the propedeuse diplom, have finished at least 30 ECTS second year courses including all practical courses, achieved excellent grades and have e.g. gained additional experience as teaching assistant. Students will be selected to enter the minor program. Contact your study advisor for approval and the supervisor of DREAM Team (Dr. ir. Wim Thijs) to set up a complete minor program of 30 ECTS. iGEM may also be used to build a free minor; please contact your study advisor to help arrange this.
Studyload/Week	Part time in Q3 Full time in Q4 and during the summer
Schedule	January, February: Team formation February April: Brainstorming meeting, Fund raising April October: Lab- & Modelling work Mid November: iGEM world-finals (Boston, USA)
Location	Delft, mainly in the Applied Sciences Building (nr. 58)

LM3771	Protein Engineering	3
Responsible Instructor	Dr. F. Hollmann	
Instructor	Dr. F. Hollmann	
Contact Hours / Week x/x/x/x	0/2-4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Elective course of the MSc programme LST.	
Expected prior knowledge	Organic Chemistry (LB1045 or 4051OCSTRY) Chemical Biology (LB2621) OR Catalysis (4052KATALY) Advanced Biocatalysis (LM3731) (or equivalent)	
Course Contents	Protein engineering is an inevitable tool for biocatalysis. Frequently, the wild-type enzymes do not meet the requirements of the desired reaction due to limitations in stability, activity or unsatisfactory selectivity. The key steps in protein engineering including (1) generation of diversity and (2) screening of the diversity to select the most suitable variant. Repetition of these steps is generally performed until the desired trait has been obtained. In this course, the basic principles of protein engineering will be discussed: (1) Generation of diversity through random mutagenesis and or gene shuffling and (semi-)rational protein design (based on e.g. crystal structures or homology models) (2) Analysis of the newly generated diversity with various analytical techniques. In particular: a. Spectrophotometric assays (e.g. NAD(P)H-based UV assays, chromogenic probes) b. Chromatographic assays (gas chromatography, high performance liquid chromatography, electrophoresis) c. NMR- and MS-based assays Furthermore, the pros and cons of the different methods will be discussed (throughput vs precision) as well as strategies to design and select a suitable assay (you get what you screen for).	
Study Goals	After attending this course the student will: - know the basics of the most common protein engineering approaches; - know the most important spectrophotometric and chromatographic screening methods; - be able to choose the most promising protein engineering- and screening method; - be able to critically evaluate the quality of a given strategy; - be able to design new screening strategies.	
Education Method	Lectures - group work - assignments	
Literature and Study Materials	Lecture Sheets, specialized books, articles and patents	
Books	suggested reading: U.T. Bornscheuer et al: Engineering the third wave of biocatalysis Nature, 485, 7397, 185-194. M.T. Reetz: Controlling the enantioselectivity of enzymes by directed evolution: Practical and theoretical ramifications Proc. Natl. Acad. Sci. USA, 2004, 101, 5716-5722.	
Assessment	(1) Written exam (digital off-campus (open book) or on-campus, depending on the COVID-19 measures), 75% of the overall grade (2) Group assignment (3-4 students describing a topic of protein engineering), 25% of the overall grade	

Year	2022/2023
Organization	Applied Sciences
Education	Master Life Science and Technology

Projects 2022

Introduction 1

INTERNSHIPS:

The following internship codes are available for students in the 2021/22 study programme.

- LM3803: Company Internship 15 EC,
- TUD4040: An alternative for the company internship is the participation in the Joint - Interdisciplinary Project 15 EC
- LM3804: In EXCEPTIONAL cases the program coordinator may allow a student to carry out a non-industrial internship, 15 EC

For students who started the study programme before the academic year 2021/22, the following internship codes are available.

- LM3802 Company Internship 18 EC
- TUD4040 An alternative for the company internship is the participation in the Joint Interdisciplinary Project 15 EC, supplemented with one elective of min. 3 EC, preferably the elective course AS3121 = Scientific writing & argumentation 3 EC
- LM3981 During covid-19 measures: in EXCEPTIONAL cases the program coordinator may allow a student to carry out an alternative project, Special Topic MSc LST 18 EC.

MASTER's THESIS PROJECT:

- LM3901 - regular Master's End Project - 45 EC
- LM3902 - code must be used in case of a Scientific & Social Orientation - 42 EC
- LM3995 - code can be used in case of Double Degree programme with a MEP of 30 EC.

LM3803	Company internship	15
Responsible Instructor	Prof.dr. U. Hanefeld	
Instructor	L. van der Elst	
Instructor	Prof.dr. U. Hanefeld	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4 Summer Holidays	
Start Education	1 2 3 4 5	
Exam Period	none	
Course Language	English	
Course Contents	You will be working full-time in a Life Science & Technology related company for a period of 3 MONTHS. Your Master Life Science & Technology knowledge is necessary for the project. The company internship is guided by an internal (TU Delft) and an external (company) supervisor. The mark for the internship is awarded by these two supervisors. The mark is based on a series of criteria, including an individual report of the student, in which the experiences concerning the main goals, including the experience in the working environment, are discussed. During the company internship you are expected to get to know the organization. To do so, you need to start gathering information about the organization of your choice well in advance, and make a list of questions you would like to have answered. During the internship you need to make time to get acquainted with other departments within the organization and interview, for example, a number of people in different parts of the organization. NOTE! LST-STUDENTS WHO STARTED 1st SEPTEMBER 2021 OR THEREAFTER, ARE ALLOWED TO REPLACE THE COMPANY INTERNSHIP (LM3803, 15 EC) BY PARTICIPATION in TUD4040 Joint Interdisciplinary Project (15 EC). In case the start of your programme was before the 1st of September 2021, you can also replace your company internship (LM3802, 18 EC) by participation in TUD4040, supplemented with one elective course of min. 3 EC, preferably AS3121 = Scientific writing & argumentation.	
Study Goals	The goal of the internship is: To become familiar with a professional working environment for a biotechnologist To use academic knowledge and skills, acquired in the degree course, to solve problems or be active in design related to Life Science & Technology, in a professional working environment. Skills should be acquired on the following topics: 1. Performance: Productivity, Independence, Practical skills, Theoretical knowledge, Creative thinking, Time management 2. Communication: Reporting, Presentation, Interpersonal 3. Competences: Motivation & enthusiasm, Critical attitude, Taking initiative	
Education Method	Practical training on the job, the company supervisor is responsible for day-to-day supervision. During the internship period the student will also report to the TU Delft instructor 2-3 times about progress made on the project.	
Literature and Study Materials	The Brightspace page http://brightspace.tudelft.nl >catalog>search for "Internship Office Applied Sciences" (enrol!) provides more information about the company internship. On this page you can find the step-by-step guide explaining all procedures, forms, internship offers, experiences from other students and additional information e.g. about grants and funding.	
Assessment	You can contact the Internship Office through: InternshipOffice-TNW@tudelft.nl For the assessment you need to submit two reports: A scientific report using the format and guidelines of the company An evaluation report Even if duration of the internship is slightly longer, these reports should be submitted and assessed after three months, as a) all students should be assessed equally, b) the assessment criteria are mainly based on a range of transferable skills and not specifically on scientific output. Two weeks before the end of your internship (3 months), the Internship Office TNW will send your company supervisor and TU Delft instructor the internship review form. Timely submission of the final report, 3 months after the starting date, is an assessment criterion as well. If this is not the case, points will be deducted. Your company supervisor needs to assess your internship at the end of your internship period and send his review form to the TU Delft instructor. Your TU Delft instructor needs to assess your internship within two weeks after the end of your internship and give a mark. The TU Delft instructor will then send the assessment form (including the company supervisor review form) to the Internship Office TNW.	

LM3804	Non-industrial Internship project	15
Responsible Instructor	Ir. A.A. Witkam	
Responsible Instructor	Prof.dr. U. Hanefeld	
Contact Hours / Week x/x/x/x	none	
Education Period	1 2 3 4 Summer Holidays	
Start Education	1 2 3 4 5	
Exam Period	none	
Course Language	English	
Course Contents	In some cases, a student may be allowed to do an internship in an environment other than industry. The learning objectives and assessment criteria are the same as those of the industrial internship (LM3802 - 18 EC, LM3803 - 15 EC). THE PROJECT MUST BE APPROVED BY THE PROGRAMME COORDINATOR OF THE DEGREE PROGRAMME. PLEASE REFER YOUR REQUEST TO Info-lst@tudelft.nl. What is allowed? - a trajectory at De Kleine Consultant (https://www.dekleineconsultant.nl/) - a trajectory at The Young Advisory Group (https://youngadvisorygroup.nl/?lang=en) - participation in the BISC-E competition (https://bisc-e.eu/) - in exceptional cases, an internship at a not-only-for-profit institute is allowed, provided that the internship is non-academic and that the institute is partially privately funded. You will be working full-time on a LST related subject for a period of 3 MONTHS. Your Master Life Science & Technology knowledge is necessary for the project. The project is guided by an internal (TU Delft) and an external supervisor. The mark for the internship/competition is awarded by these two supervisors. The mark is based on a series of criteria, including an individual report of the student, in which the experiences concerning the main goals, including the experience in the working environment, are discussed. During the internship/competition you are expected to get to know the organization. To do so, you need to start gathering information about the organization of your choice well in advance, and make a list of questions you would like to have answered. During the internship you need to make time to get acquainted with other departments/parties/fellow students within the organization and interview, for example, a number of people in different parts of the organization/institute.	
Study Goals	The goal of the internship is: - To become familiar with a professional working environment for a biotechnologist - To use academic knowledge and skills, acquired in the degree course, to solve problems or be active in design related to Life Science & Technology, in a professional working environment. Skills should be acquired on the following topics: 1. Performance: Productivity, Independence, Practical skills, Theoretical knowledge, Creative thinking, Time management 2. Communication: Reporting, Presentation, Interpersonal 3. Competences: Motivation & enthusiasm, Critical attitude, Taking initiative	
Education Method	Practical training on the job, the external supervisor is responsible for (day-to-day) supervision. During the internship period the student will also report to the TU Delft instructor 2-3 times about progress made on the project.	
Literature and Study Materials	The Brightspace page http://brightspace.tudelft.nl >catalog>search for "Internship Office Applied Sciences" (enrol!) provides more information about the company internship. On this page you can find the step-by-step guide explaining all procedures, forms, internship offers, experiences from other students and additional information e.g. about grants and funding.	
Assessment	You can contact the Internship Office through: InternshipOffice-TNW@tudelft.nl For the assessment you need to submit two reports: - A scientific report using the format and guidelines as published on the Brightspace organization Internship Office Applied Sciences - An evaluation report Even if duration of the internship is slightly longer, these reports should be submitted and assessed after three months, as a) all students should be assessed equally, b) the assessment criteria are mainly based on a range of transferable skills and not specifically on scientific output.	

LM3901	Master Thesisproject	45
Responsible Instructor	Ir. A.A. Witkam	
Responsible Instructor	Prof.dr. U. Hanefeld	
Instructor	Dr. W.M. van Gulik	
Instructor	Prof.dr.ir. P.A.S. Daran-Lapujade	
Instructor	Dr. F. Hollmann	
Contact Hours / Week x/x/x/x	Based on appointments with your supervising tutor	
Education Period	None (Self Study)	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The student has to work independently on a research project. This project is specialization-specific. The work deals with the following aspects: - Literature study, problem formulation and planning - Practical and theoretical work in one of the research groups, including interpretation and evaluation of the results - Oral and written presentation of the work	
Study Goals	After the Master Thesis Project the student has shown to be able: - to read and understand international literature of a specific topic - to work independently on an academic level in a R&D environment. Planning is a keyword! - to work in an interdisciplinary and multicultural team of experts and supporting technologists/analysts - to present his/her results in English by means of a presentation to a professional audience and by means of a written scientific report	
Education Method	Individual Project carried out within a Research Group	
Literature and Study Materials	Provided by the research group	
Assessment	Student has to write, present and defend a thesis. Assessment criteria focus on - Scientific knowledge & Understanding - Method & approach - Report - Presentation & Defense - Competences Assessment criteria are laid down in an assessment matrix --> see Brightspace Master Life Science & Technology --> content --> study info --> Thesis project. IMPORTANT REMARK: This grading scheme is part of the digital review form that will be filled out by your thesis supervisor. The digital review form concludes with 4 separate questions and corresponding tick boxes: The student is sufficiently aware of (i) safety, (ii) data safety, (iii) environmental and (iv) ethical regulations and respects these in his/her work. In case the safety criteria are not met, student will not successfully pass the thesis project.	
Enrolment / Application	Students have to send in a MEP application form. See details on Brightspace: - Master Life Science & Technology -> content -> study info -> MSc Thesis Project - Thesis Office Applied Sciences	
Special Information	IMPORTANT INFORMATION: Special course codes are created for - students with a double degree thesis project of 30 EC: LM3995 - students with a 30 EC 'Scientific and Social Orientation' in their exam programme, perform a thesis project of 42 EC -> LM3902	
Remarks	Certain steps need to be taken when carrying out a Thesis Project. Please, consult Brightspace (Thesis Office Applied Sciences and or Master Life Science & Technology) for detailed information and additional forms. The procedure of a Thesis Project consists of the following steps: 1. Orientation 2. Careful consideration of different research sections. Make an appointment with your specialization coordinator, before or shortly after your search for a project, and discuss the options. 3. Handing in the application form and a list of the preferred elective courses. MAKE SURE THAT YOU AGREED ON A PROPER PLANNING WITH THE THESIS SUPERVISOR (the application form goes with planning guidelines) 4. Providing the Thesis Project Administration with the names of the review committee. Note: it's the responsibility of the thesis supervisor to compose the review committee. 5. Presentation 6. Assessment and calculation of the examination mark !! Pay attention !! The final mark will only be registered after the Thesis Project Administration has received a digital copy of the thesis report. A digital survey will be sent to the student shortly hereafter. For questions & handing in the digital version of the report, contact ThesisOffice-TNW@TUDelft.nl. In exceptional cases (Double Degree) the Board of Examiners may decide to limit the duration and the consequent number of credits, in good agreement with the specialization coordinator.	

TUD4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Project Coordinator	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Summary	JIP consists of three sets of activities: 1. The project which takes place at a company. Students are responsible for the project work (collaboration, integration of different perspectives, the content quality of their work). They plan their activities via a scrum method, are challenged to realise field trips, consult with experts and realise the necessary research work. The team keeps a Scrum log, run a blog for interaction with the outside world and every team member keeps a personal development log. 2. Meeting Professionals lectures and workshops about specialised topics, by academic staff or senior professionals from the companies involved. 3. Plenary meetings and design reviews for all the teams, company coaches and academic staff. At the meetings the students present their intermediate resp. final outcomes and plans, and are provided with feedback.	
Course Contents	The aim of the Joint Interdisciplinary Project is to prepare students to contribute to solving impactful technological challenges. The projects not only demand good engineering working knowledge but also experience with interdisciplinary and systems theory, and both knowledge and mindsets of innovation and entrepreneurial behavior. The project brief is provided by renowned companies like Airbus, Arcadis, etc. Teams of interdisciplinary student teams guided by a company coach and offered academic and industry expertise, are invited to realise an innovative problem solution to a complex problem and contributing to the sustainable development goals.	
Study Goals	1. Cognitive abilities attributable to interdisciplinary learning Demonstrate the ability to engage in perspective-taking; Develop structural knowledge pertaining to the problem; Integrate knowledge and modes of thinking drawn from two or more disciplines; Produce an interdisciplinary understanding of complex problem or intellectual question. 2. Scientific and intellectual development Capable to analyze scientific and societal consequences (economic, social, cultural, environmental) of the innovation; 3. Research and design capabilities Demonstrate engineering skills: technical skills, interpreting results, creativity, usability for company/institute; Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; 4. Collaboration and communication in an interdisciplinary team Demonstrate behavioral competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to respect different disciplines and adapt to different cultures); Show ability to write a technical report: structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (30 pages); Present work performed in a structured way through an oral presentation to their peers and customer. 5. Self-adjustment and reflection capabilities Plan and control the project efficiently considering resources and methodology; Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career. Cognitive abilities attributable to interdisciplinary learning: The ability to integrate (scientific and practical technological) knowledge from different disciplines to solve complex problems Scientific and intellectual development The capacity to evaluate the ethical, scientific and societal consequences of the proposed innovation Research and design capabilities The ability to create reasonable and relevant research or design , according to the academic standards of the involved disciplines Collaboration and communication in an interdisciplinary team Demonstrate behavioural competences and skills relevant for teamwork and effective communication with different stakeholders Self-adjustment and reflection capabilities To carry out regular reflections on professional and personal development and being able to improve upon those reflections Understand contemporary and societal issues in their work.	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	The assessment criteria pertain to the process, product and presentation and are assessed at an individual and team level, during 3 review sessions at three levels of accomplishment (each review assesses at a different level): Interdisciplinary work Scientific reasoning and ethical mindset Innovation process Presentation and communication Academic staff, supported by the input from company coaches, grade the students during the Problem Statement, the Midterm and the Final Review. The final group mark for JIP is differentiated per person and is based on: The team presentations - 30% The Problem definition, progress report and final report 30% The individual contribution to the team 20% The final individual reflection report 5% The Blog 5%	
Remarks	Project Outcomes (product, model, system) are presented at: 1. Problem Statement Review 2. Midterm meeting 3. Final Review 4. Symposium presentation	

Year	2022/2023
Organization	Applied Sciences
Education	Master Life Science and Technology

Specialisation Modules 2022

Year	2022/2023
Organization	Applied Sciences
Education	Master Life Science and Technology

Cell Factory 2022

Introduction 1

The specialisation 'Cell Factory' focuses on the design, understanding and optimization of living cells as environmentally and economically sustainable production systems. The courses in this track focus on acquiring and integrating the scientific knowledge needed to use (microbial) cells and communities for the production of valuable components ranging from food ingredients to fuels. The embedding of this specialisation in the Delft Life Science and Technology MSc programme ensures that students joining this Cell Factory specialisation will learn how to integrate and communicate knowledge on fast-moving areas such as systems biology, synthetic biology and community engineering of industrial microorganisms together with the specialist knowledge bioprocess engineers and specialists in biocatalysis. The combination of scientific depth, attention for industrial application and engineering aspects offer a unique perspective for a further career in academic or industrial research.

LM3432	Analysis of Metabolic Networks	6
Responsible Instructor	Dr. W.M. van Gulik	
Course Coordinator	Dr. W.M. van Gulik	
Instructor	T.W. Páez Watson	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The focus of AMN is on the quantitative analysis and understanding of living cells and their interaction with the environment. Especially, specific rates (growth, production) which result from genetic properties and a highly connected set of networks: - Metabolic reaction networks - Genetic regulation networks During the course we will discuss different modelling approaches to evaluate and predict cellular metabolism. This includes prediction of the effect of genetic and environmental changes on rates in biological systems. Modeling is useful to test hypothesis and understand observed behavior. For a successful modeling cycle, the connection with the experiments is essential - the course addresses experimental methods to quantify fluxes in networks We will present three levels of detail in the course: - Black box kinetics & thermodynamics (refreshment of Bachelor courses and some new) - Medium scale stoichiometric models - Small scale kinetic models Further the course contains: - Refreshment of prior knowledge (mass balances, conversion rates, black box kinetics/ stoichiometry, energy-, product-, growth pathways) - Validation of experimental data & data reconciliation - Metabolic flux analysis of large networks - Transport over cellular membranes - Kinetics of metabolic pathways - Mathematical modeling of central carbon metabolism - Regulation of enzyme expression, principles & modeling - Sources of bio-information/ databases	
Study Goals	To perform the calculations, python is used as simulation tool for biological processes. After the course the student should be able to - Set-up massbalances for bioreactor processes - Use linear algebra to estimate and reconcile metabolic rates - Relate metabolic network structure to maximal biomass and product yields/ product rates - Show understanding of the thermodynamic and redox constraints in metabolic networks - Define experiments to calculate fluxes - Set-up mathematical models which relate network/ enzyme properties to fluxes - Interpret results from stoichiometric and kinetic modeling to define engineering targets. Most relevant transferable skills - Communication - Problem solving - Teamwork - Time management	
Education Method	Lectures and exercise classes	
Assessment	Group assignments (50%), transferable skills acquired are teamwork, communication, time management, problem solving Final individual exam (50%)	
Special Information	For the 3rd course of their specialization Biochemical Engineering, students can make a choice out of LM3611 Microbial Community Engineering LM3432 Analysis of Metabolic Networks	

LM3442	Metabolic Reprogramming	6
Responsible Instructor	Prof.dr.ir. P.A.S. Daran-Lapujade	
Instructor	Prof.dr.ir. P.A.S. Daran-Lapujade	
Contact Hours / Week x/x/x/x	0/3/4-8/0	
Education Period	2 3	
Start Education	2	
Exam Period	3 4	
Course Language	English	
Required for	MSc specialisation Cell Factory, Life Science & Technology	
Expected prior knowledge	This course requires BSc-level knowledge in Microbial Physiology, Molecular Genetics and insight into the quantitative aspects (yields, rates) of Biotechnological processes (LST BSc courses LB2762TU, LB2161 and LB1511TU or equivalents thereof).	
Parts	Lectures schedule: 1. General introduction to metabolic reprogramming and the course First lecture on increasing fluxes and yields in existing pathways 2. Classical strain improvement and reverse engineering 3. Evolutionary and reverse engineering 4. Synthetic biology, new products 5. Heterologous protein production Journey into the publication process 6. Mid-term exam and collegial correction	
Course Contents	Application of targeted genetic modification and rational application of evolutionary strategies for the optimization of existing product formation pathways in micro-organisms, introduction of new product formation pathways, physiological constraints on process optimization	
Study Goals	After this course, the students should: - understand fundamental aspects, possibilities, limitations and challenges in the rational optimization of microbial cells for industrial product formation - be able to critically read scientific papers and discuss their contents with colleagues in a structured manner	
Education Method	This course is a combination of a lecture series and presentations/discussions of scientific publications. The lecture series will consist of 6 three-hour sessions and will introduce key principles of metabolic reprogramming and introduce the format of the second part of the course. For this second part, a collection of relevant scientific publications will be handed out. Each student participating in the course prepares a critical (oral) presentation on one of these papers. Furthermore, all students read the complete set of publications, which allows them to participate in plenary discussions (led by the teachers) on the papers. In addition to the scientific content, attention will be given to key quality aspects of the scientific publications (style, statistics, internal consistency, impact). Sessions of 2 hours will be used for the presentation and discussion of two scientific papers. Education and exams will be on site.	
Literature and Study Materials	A set of scientific publications will be provided by the teachers at the outset of the course (via Brightspace)	
Assessment	Students will be assessed based on: 1. A short mid-term exam which goal is to assess student's ability to critically analyse scientific literature. (10% of final grade) 2. Their presentation and critical analysis of a chosen scientific publication. (25% of final grade) 3. Their critical analysis of a chosen article and scientific discussion with the group (called Critical Panel, 20%) 4. Their contribution to the discussion following presentations by their fellow participants. (20% of final grade) 5. A final, written exam that will cover the subject matter taught in the lecture series as well as key elements discussed during the evaluation of the scientific paper. (25% of final grade) While students are encouraged to analyse together scientific literature, assessment of the course is purely based on individual performance. Resits in period 4 by written or oral examination.	
Permitted Materials during Tests	Pen, paper, calculator, annotated article (supplied in advance)	
Enrolment / Application	not later than one week before the first lecture via BrightSpace	
Location	Department of Biotechnology, Delft University of Technology	

LM3601	Molecular Biotechnology & Genomics	6
Responsible Instructor	Prof.dr.ir. J.G. Daran	
Instructor	Prof.dr.ir. J.G. Daran	
Contact Hours / Week x/x/x/x	0/3/4/0	
Education Period	2 3	
Start Education	2	
Exam Period	3 4	
Course Language	English	
Course Contents	The Molecular Biotechnology and genomics course offers an in-depth overview of contemporary biotechnology. The program aims at providing the student with a profound fundamental basis, as well as offering him an integrated study package, compiled from a wide scope of scientific disciplines, which together underscore the practical realizations of application-oriented molecular biotechnology.	
Course Contents Continuation	1. Genomes (strains, genome sequences, sequencing technologies,). Origin of industrial strains/applications Sequencing industrial organism, why? Methods from Sanger to next generation sequencing techniques Analysis of genome sequences 2- Strain improvement strategies. Classical strain improvement High throughput screening genome shuffling Molecular Genetics/Metabolic engineering Examples 3- Molecular genetics (strain design) Genetic engineering solutions Deletion, silencing, overexpression Vector design Restriction versus recombination) Host design integration HR vs NHEJ 4 Cloning strategies Identification of industrial enzymes by expression cloning 5-Transcriptomics. 6- Computer classes Plasmid design reference guided and de novo genome assembly	
Study Goals	After this course the students should: Obtain a deeper understanding of the possibilities and limitations of molecular cell biology/molecular genetics and genomics applications in modern biotechnology. Integrate knowledge on (i) molecular cell biology/molecular genetics, (ii) genomics and (iii) cell factory research.	
Education Method	E.G.: Lecture series and hands-on exercises Computer classes to acquire hands on experience. This course is a combination of a lecture series and in-silico practical exercises The course is divided in two parts: in the first a lecture series accompanied with relevant exercises will introduce key concept in molecular genetics. The second part will be based on do it yourself genomics classes aiming at teaching principles of genome data, assembly analysis. The computer sessions will be punctuated with explanatory lectures.	
Assessment	Students are evaluated by 3 different methods. 1. Two individual assignments are proposed to the students during the course, two in the first period and one in the second. The students have between 2 to 3 weeks to complete it and to give it back to teacher before a clearly defined deadline. Each assignment accounts for 20% of the final grade. 2. Time limited (3h30) assignment on genome assembly. Due the complexity of the exercise, this assignment needs to be tightly supervised to allow the students to progress. The grade of this assignment is used as a bonus (up to 10% of the final written exam grade) added to the grade of the final written exam. 3. Written exam (60% of final grade of the final grade). All grades are individual	
Enrolment / Application	not later than one week before the first lecture via Brightspace Delft (courses > Master > LS&T)	
Location	To be mentioned on due course. See also: www.timetables.tudelft.nl	

Year	2022/2023
Organization	Applied Sciences
Education	Master Life Science and Technology

Biocatalysis 2022

Introduction 1

The specialisation 'Biocatalysis' integrates enzymological, bio-organic, bio-inorganic and protein-analytical knowledge, with the aim to understand the principles of biocatalysis, and provides theoretical and practical guidelines to the determination of structure and function of single enzymes or multiple enzymes (proteomics/MS). In particular, the students will be familiarized with modern kinetic, analytical and computer graphical methods used in present day protein/peptide/enzyme research. As catalysts, enzymes are essential proteins in all living organisms and are being applied in numerous industrial conversions. This specialisation gives the necessary background how to deal with enzymes in practice, how to study their properties, structure, function and possible relation to disease, and how to choose the most suitable enzyme for a specific industrial application.

LM3434	Advanced Protein Chemistry and Analysis		6
Responsible Instructor	Dr. M. Pabst		
Responsible Instructor	Dr. D.G.G. McMillan		
Responsible Instructor	Dr. K. Djanashvili		
Module Manager	Dr. K. Djanashvili		
Module Manager	Dr. M. Pabst		
Module Manager	Dr. D.G.G. McMillan		
Contact Hours / Week x/x/x/x	0/0/6/0		
Education Period	3		
Start Education	3		
Exam Period	3		
Course Language	English		
Expected prior knowledge	Biochemistry, Chemistry, Basic NMR principles		
Course Contents	Proteins are the engineers of the cell, performing roles from shaping the cell to complex catalysis and energy production. In doing this, they interact with one another and with other biological macromolecules, forming complex assemblies. However, protein function relies not only on the amino acid sequence, but also on covalent modifications capable of enhancing or inhibiting protein activities. We also explore how proteins and modifications are identified and characterised using cutting-edge NMR spectroscopy, high performance chromatography and high-resolution mass spectrometry. This course will provide a solid foundation to the protein structure and function		
Course Contents Continuation	Course layout Lecture 1: Protein structure/function and the cell D.McMillan Lecture 2: Complex Structures and Common Motifs D.McMillan Lecture 3: Protein synthesis, Folding, and Basic engineering D.McMillan Lecture 4: Advanced Engineering and Enzyme Design D.McMillan Lecture 5: Methods in Protein Structure D.McMillan Lecture 6: Article Cross-Examination D.McMillan Lecture 7: Concepts of covalent protein modifications M.Pabst Lecture 8: Important protein modifications in nature M.Pabst Lecture 9: NMR protein structure K.Djanashvili Lecture 10: NMR interactions/applications K.Djanashvili Lecture 11: Protein purification and characterisation I M.Pabst Lecture 12: Protein purification and characterisation II M.Pabst Lecture 13: Native MS M.Pabst Question Workshop D.McMillan, K.Djanashvili, M.Pabst		
Study Goals	Students will be able to apply basic protein structure principles to protein function and enzyme catalysis. Students will be able to evaluate protein design approaches and apply the knowledge to design enzymes for optimal enzyme specific activity and yield. Students will be able to identify the defining principles behind post-translational modification and the purpose behind them Students will be able to evaluate data from various protein analytical techniques and be able to identify appropriate.		
Education Method	Lectures, Tutorials and Peer-Assessment		
Literature and Study Materials	Hand-outs		
Books	Recommended Reading Branden and Tooze, Introduction to Protein structure 2nd Edition. ISBN-10:0815323050 Introducing Proteomics: From Concepts To Sample Separation, Mass Spectrometry and Data Analysis. WILEY books, Josip Lovric. Omics Technologies and Bio-Engineering: Chapter 15. 2017, Debmalya Barh and Vasco Azevedo. Post-Translational Modifications of Proteins: Christoph Kannicht, 2002, Springer books. Post-Translational Modifications of Proteins: Expanding Nature's Inventory; Christopher T. Walsh.		
Assessment	Individual Assignment (20%) and final written exam (80%)		

LM3701	Advanced Enzymology	6
Responsible Instructor	Dr.ir. P.L. Hagedoorn	
Instructor	Dr.ir. P.L. Hagedoorn	
Instructor	Dr. D.G.G. McMillan	
Contact Hours / Week x/x/x/x	0/4/3/0	
Education Period	2 3	
Start Education	2	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	BSc LST, BSc MST or equivalent in terms of Biochemistry/Chemistry	
Course Contents	The course Advanced Enzymology covers basic biochemical principles, which are applied to understanding catalysis and energy conversion by (metallo-)enzymes, their biological function and their dysfunction in disease. This course also focuses on theoretical and practical backgrounds of various spectroscopic, electrochemical and pre-steady state kinetic methods, as well as on the use of computer visualization programs and the enzymology of membrane enzymes. Students will learn how to employ these methodologies to the determination of the structure of metallo-centers and function in terms of the molecular mechanism of catalysis. To successfully apply metalloenzymes in research or industrial applications, specific problems related to the (over)expression of metallo-proteins such as cofactor biosynthesis - are addressed. All these aspects will be illustrated by using two important enzymes: cytochrome P450 and methane mono-oxygenase.	
Study Goals	After the course the students should - be able to apply basic bio-inorganic principles to metallo-enzyme catalysis, understand the biological role of metallo-enzymes and the consequences of their dysfunction in disease - have obtained a basic theoretical level in various spectroscopic and kinetic methods, in electron transfer theory and be able to apply theory to the determination of metallo-enzyme structure and molecular mechanism - be able to judge the scope and limitations of metallo-enzymes in industrial applications - be able to design (metallo)enzyme production, isolation and assay procedures and visualize 3D enzyme structures - compose a strategy for the design of a novel biocatalyst for a desired chemical reaction - use theoretical knowledge of lipid membranes to understand the mechanism of membrane transport proteins and membrane enzymes - show responsibility to complete an individual assignment on time with sufficient effort - able to independently solve a scientific problem related to advanced enzymology	
Education Method	The course is given as a series of live (online or on-campus) lectures of 3 hours (6 times) and 4 hours (6 times). After each lecture a formative assignment will be handed out as a self-assessment assignment on Brightspace, which will be discussed during the subsequent lecture. The course includes a 4 hour computer practical on enzyme structure visualization and analysis using PyMOL software. During the course the students will also read recent scientific papers and discuss the contents.	
Literature and Study Materials	Syllabus and lecture slides	
Assessment	Written exam (digital off-campus or on-campus, open book; 2/3 of final grade) and an individual assignment (1/3 of final grade). The following transferable skills will be included in the assessment of the individual assignment: planning and problem solving (see study goals).	
Enrolment / Application	not later than one week before the first lecture via Brightspace Delft (courses > Master > LS&T)	
Schedule	Metals in biology - Introduction bio-inorganic chemistry - Ligand field theory and spectroscopy - Biosynthesis of metallo-cofactors - O₂ in biology Bioelectrochemistry and Bioenergetics - Electron transfer and redox catalysis - Mitchell hypothesis and theory Enzymology and enzyme kinetics - Enzymes: sources, isolation, assays - Steady-state kinetics - Pre-steady state kinetics - PyMol visualization of 3D enzyme structures - Computational enzymology/ de-novo enzyme design - Membrane enzymology	

LM3731	Advanced Biocatalysis	6
Responsible Instructor	Prof.dr. U. Hanefeld	
Module Manager	Prof.dr. U. Hanefeld	
Module Manager	C.E. Paul	
Contact Hours / Week x/x/x/x	0/4/3/0	
Education Period	2 3	
Start Education	2	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	BSc degree LST or equivalent	
Course Contents	This course gives an introduction to the use of enzymes for the synthesis of chemicals and pharmaceuticals. Starting from a basic knowledge in biochemistry, enzymology and organic chemistry, we will study proteins for their capacity to act as catalysts for different classes of reactions (such as reductions, oxidations and C-C bond formations). The importance of enzymes for the synthesis of chiral compounds will be an important theme. In addition the advantage and technique of immobilization of enzymes will be treated.	
Study Goals	After this course the student should have acquired in depth knowledge about enzymes as biocatalysts and their application in Organic Chemistry. Based on this knowledge the student should be able to choose a suitable biocatalyst for an industrially important reaction and he/she should be able to explain why which type of approach is more suitable for the production of a given bulk or fine chemical. This includes an in depth analysis of the enzyme, the working conditions and the mechanism. Moreover, the students will acquire presentation skills via a structured preparation and presentation of a lecture on a self chosen topic.	
Education Method	Lectures, Self-study and a lecture by the student	
Literature and Study Materials	Handouts/Brightspace Books: K. Faber, Biotransformations in Organic Chemistry, 7th ed., Springer-Verlag, Berlin, 2018. available via the TU library - Soderberg, Timothy, "Organic Chemistry with a Biological Emphasis Volume I" (2019). Chemistry Publications. 1. https://digitalcommons.morris.umn.edu/chem_facpubs/1 - Soderberg, Timothy, "Organic Chemistry with a Biological Emphasis Volume II" (2019). Chemistry Publications. 2. https://digitalcommons.morris.umn.edu/chem_facpubs/2	
Assessment	Literature studies in the lecture theatre. Written examination (80%, digital off campus (open book) or on campus (no open book), depending on COVID-19 campus regulations), Presentation skills will be examined as part of a review and presentation of a self-chosen scientific paper (20%). Resit period 4, 2022-2023: oral, on request.	
Enrolment / Application	not later than one week before the first lecture via Brightspace (courses > Master > LS&T).	
Schedule	1. Introduction 2. Why are enzymes such good catalysts/kinetics 3. Classes of enzymes 4. Hydrolysis/enantioselectivity 5. Reaction media: water or no water 6. C-C bond formation 7. Reductions 8. Oxidations 9. Immobilisation 10. Other relevant enzyme classes 11. Comparison of the enzymes and their application.	
Location	Delft	

Year	2022/2023
Organization	Applied Sciences
Education	Master Life Science and Technology

Biochemical Engineering 2022

Introduction 1

Bioprocess Engineering is concerned with new processes that involve cells or parts thereof such as enzymes, and which are based on sustainable feedstocks. Various application fields rely on bioprocess engineers. For example, production of vitamins, drugs, vaccines, diagnostics, bioplastics, and biofuels; manufacture of foods and drinks; cleaning of wastewater and recycling of waste streams. Students learn how to design, and how the engineering aspects of bioprocesses call for scientific understanding of the underlying cellular, molecular and physical principles. Available basic knowledge is integrated and often supplemented by own research in engineering and biotechnology. A quantitative approach is used, involving mathematical modelling and computer tools. Therefore assessment of many courses is by assignments.

NOTE:

LM3741 Fermentation Technology & Environmental Biotechnology - compulsory

LM3751 Transport & Separation - compulsory

Choice out of the 2 courses: LM3611 or LM3432

LM3611 Microbial Community Engineering

LM3432 Analysis of Metabolic Networks

LM3432	Analysis of Metabolic Networks	6
Responsible Instructor	Dr. W.M. van Gulik	
Course Coordinator	Dr. W.M. van Gulik	
Instructor	T.W. Páez Watson	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The focus of AMN is on the quantitative analysis and understanding of living cells and their interaction with the environment. Especially, specific rates (growth, production) which result from genetic properties and a highly connected set of networks: - Metabolic reaction networks - Genetic regulation networks During the course we will discuss different modelling approaches to evaluate and predict cellular metabolism. This includes prediction of the effect of genetic and environmental changes on rates in biological systems. Modeling is useful to test hypothesis and understand observed behavior. For a successful modeling cycle, the connection with the experiments is essential - the course addresses experimental methods to quantify fluxes in networks We will present three levels of detail in the course: - Black box kinetics & thermodynamics (refreshment of Bachelor courses and some new) - Medium scale stoichiometric models - Small scale kinetic models Further the course contains: - Refreshment of prior knowledge (mass balances, conversion rates, black box kinetics/ stoichiometry, energy-, product-, growth pathways) - Validation of experimental data & data reconciliation - Metabolic flux analysis of large networks - Transport over cellular membranes - Kinetics of metabolic pathways - Mathematical modeling of central carbon metabolism - Regulation of enzyme expression, principles & modeling - Sources of bio-information/ databases	
Study Goals	To perform the calculations, python is used as simulation tool for biological processes. After the course the student should be able to - Set-up massbalances for bioreactor processes - Use linear algebra to estimate and reconcile metabolic rates - Relate metabolic network structure to maximal biomass and product yields/ product rates - Show understanding of the thermodynamic and redox constraints in metabolic networks - Define experiments to calculate fluxes - Set-up mathematical models which relate network/ enzyme properties to fluxes - Interpret results from stoichiometric and kinetic modeling to define engineering targets. Most relevant transferable skills - Communication - Problem solving - Teamwork - Time management	
Education Method	Lectures and exercise classes	
Assessment	Group assignments (50%), transferable skills acquired are teamwork, communication, time management, problem solving Final individual exam (50%)	
Special Information	For the 3rd course of their specialization Biochemical Engineering, students can make a choice out of LM3611 Microbial Community Engineering LM3432 Analysis of Metabolic Networks	

LM3611	Microbial Community Engineering	6
Responsible Instructor	D.G. Weissbrodt	
Responsible Instructor	Prof.dr.ir. M.C.M. van Loosdrecht	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	BSc-level knowledge in Microbial Physiology (LST BSc-course LB2762TU or equivalent thereof)	
Course Contents	The course will discuss the metabolic potential of microbes and their role in the cycling of chemical elements. It will indicate the fundamental aspects of microbial competition and cooperation in natural and man-made environments. Specific attention is given to the translation of ecological insight in the development of microbial community-based bioprocesses, such as wastewater treatment, biogeochemistry, biocorrosion, and the production of chemicals and bioenergy.	
Study Goals	After this course, the students should be able to: - Describe the concept and methodology of natural selection-based microbial community engineering as a means for the development of bioprocesses. - Capture the impressive microbial diversity in nature and its potential application in biotechnology. - Process the global element cycles and the organisms that catalyze the reaction in these cycles. - Characterize and implement the different diagnostic tools that are currently available to study the diversity and activity of microorganisms in natural and man-made environments. - Formulate scientific research questions to investigate natural and technical microbiomes, develop an investigation method, and predict the distributed functional performance of the microbiome. - Write a concise scientific essay / perspective article on a selected topic in microbiome science & engineering.	
Education Method	Microbime science and enineering is a very indterdisciplinary field. The course is built on a lecture series, given by specialists of their fields. Each lecture will bring the basics and broadens on the applicability of the concepts. Besides personal development on concepts and methods, peer learning as well as critical and creative thinking will be fostered via the production of an essay / perspective article. The highlights of the essay will be presented in a poster presentation. The MCE learning achievements will be assessed on an individual written exam. T	
Literature and Study Materials	- Session handouts on Brightspace - Chapter: Weissbrodt, Laurení, van Loosdrecht, Comeau (2020) Basic Microbiology & Metabolism; in: Biological Wastewater Treatment: Principles, Modeling and Design. 2nd Edition, IWA Publishing, UK. - Optional book: Madigan, Bender, Buckley, Sattley, Stahl (2018) Brock Biology of Microorganisms, 15th Edition, Pearson Higher Edudaction, USA.	
Assessment	Learning objectives: Fundamentals of MCE Core cycles in MCE Advanced methods in MCE Concepts integration Critical thinking and scientific creativity Assessment methods: (1) Individual written examination (50% of course score): ONSITE or ONLINE EXAMINATION depending on COVID-19 situation and in-house regulations. (2) Group essay (50% of course score) >> 2.1 Group manuscript (30% of course score) >> 2.2 Group presentation (10% of course score) >> 2.3 Individual assessment in presentation (10% of course score)	
Enrolment / Application	Not later than one week before the first lecture through Brightspace (courses > Master > LS&T)	
Special Information	For the 3rd course of their specialization Biochemical Engineering, students can make a choice out of LM3611 Microbial Community Engineering, or LM3432 Analysis of Metabolic Networks	
Remarks		
Schedule	(the tentative schedule can be subjected to changes) Part 0. Introduction to Microbial Community Engineering (MCE) > MCE Introduction / Weissbrodt Part 1. Fundamentals of MCE > Microbial ecology principles / Weissbrodt > Redox reactions in microbial conversions and growth / Kleerebezem > Thermodynamics of microbial growth / Kleerebezem > Role of biopolymers in microbial ecophysiology / van Loosdrecht > Microbial competition along gradients and biofilms / Picioreanu Part 2. Core Cycles and Conversions in MCE > Nitrogen bioconversion network / Laurení > Metabolic puzzles on the nitrogen cycle / Kleerebezem > Organic carbon cycle: anaerobic conversions and chain elongation / Kleerebezem > Phosphorus cycle and granular sludge / van Loosdrecht > Extracellular polymeric substances / Lin > Bioelectrochemical conversions / Jourdin > Sulfur cycle / Sorokin > Phototrophic conversions / Weissbrodt + Cerruti Part 3. Classical and Advanced Analytical Methods for MCE	

- > Community systems microbiology and ecophysiology / Weissbrodt
- > Metagenomics and metatranscriptomics / Weissbrodt
- > Introduction to proteomics / Pabst
- > Metaproteomics applications / Pabst
- > Metabolic flux analysis in mixed-culture systems / Wahl
- > Resource allocation to predict microbial performance / Wahl

Part 4: Concepts and Data Integration in MCE

- > Concepts and data integration in MCE / Weissbrodt
- > Numerical methods for MCE / Weissbrodt + Calderón Franco
- > MCE Conclusion / Weissbrodt

EBT research projects and lab tour / Laureni + PhDs + postdocs

Location TU Delft and/or on-line depending on public health condition and regulations

LM3741	Fermentation Technology & Environmental Biotechnology	6
Responsible Instructor	Dr. W.M. van Gulik	
Responsible Instructor	Dr.ir. R. Kleerebezem	
Instructor	Dr. W.M. van Gulik	
Instructor	Prof.dr.ir. M.C.M. van Loosdrecht	
Instructor	Dr.ir. R. Kleerebezem	
Instructor	Prof.dr.ir. H.J. Noorman	
Contact Hours / Week x/x/x/x	0/5/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Microbial diversity, Black box kinetics of growth/ product formation, mass balances, transport phenomena.	
Course Contents	The FTEB course is divided into two sections: Microorganism characterization (i), and Bioreactor design (ii). Microorganism characterization deals with stoichiometry, kinetics, and thermodynamics of microbial processes in the field of FT and EB. Specific attention is paid to identification of parameters and parameter sensitivity. The bioreactor design section includes batch, fed-batch and continuous bioreactor operation. Specific attention is paid to mixing, gas-liquid mass transfer, biofilm and other biomass retention based bioprocesses, as well as integration of different microorganisms in a single process configuration. All lectures are supplemented with practical examples. Students will elaborate in groups of 3-4 students a case study that should result eventually in the design of a bioprocess. As in the lecture series, students will first elaborate the properties of the microorganisms required and subsequently on the implementation in the bioreactor. Both parts of the case study will be discussed in classical sessions.	
Study Goals	After this course the students are able to analyse and design fermentation and environmental bioprocesses. Most relevant transferable skills: - Problem solving - Teamwork - Communication - Responsibility - Time management - Presentation skills	
Education Method	Lectures and assignments	
Literature and Study Materials	Lecture notes, lecture slides and relevant scientific papers, which will be provided via Brightspace.	
Assessment	Case study (group work) and exam (individual). The scores for the case study and the exam determine for 50% the final grade. The minimum exam score to pass is 5.	
Enrolment / Application	On Brightspace. Not later than one week before the first lecture	
Schedule	Part 1: Microorganism - Introduction - Stoichiometry - Thermodynamics - Basic kinetics - Including product formation - Determining parameters from experimental data - Fed-batch, batch, chemostat - Case study part 1: microorganisms Part 2: Bioreactor - Mass transfer modelling - Design transfer processes in bioreactors - Biofilms structure,function, flux analysis, and process design - Integrating mass transfer, biofilms, and biomass retention - FT-examples - EB-examples - Case study part 2: Bioreactor	
Location	Delft	

LM3751	Transport & Separation	6
Responsible Instructor	Dr. M.E. Klijn	
Responsible Instructor	Prof.dr.ir. M. Ottens	
Module Manager	Prof.dr.ir. M. Ottens	
Module Manager	Dr. M.E. Klijn	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	1. Introductory refreshment of transport phenomena, thermodynamics, and basic concepts of separation processes. 2. Process equipment sizing and link to thermodynamic cycles and energetic yield. 3. Multi-component separations: equilibria, heat and mass transfer in multi-component systems.	
Study Goals	Knowledge, theory and computational skills. At the end of this course, you will be able to LO1. Apply basic principle underlying separation processes for the design of separation equipment LO2. Develop Python-based models for separation equipment used in bioprocessing for simulation purposes LO3. Assess the performance of bioprocess separation steps including mass and energy balances on the basis of its efficiency LO4. Compare separation processes based on their driving force mechanism and calculated (energy) efficiency Transferable skills At the end of this course, you will be able to LO5. Breakdown and discuss computational approaches and theory used to solve a homework assignment via a PowerPoint presentation LO6. Combine problem solving skills and teamwork to complete group assignments	
Education Method	Problem-based and peer learning	
Assessment	Final grade is made of contributions from: 40% group assignments (skills: teamwork and problem solving) 10% exercises presentations (skills: communication and presentation) 50% individual written examination in Python (skills: problem solving) A minimum grade of 5 at the individual written examination is needed to pass the course.	

Scientific and Social Orientation 2022

Introduction 1

Scientific & Social Orientation

Students are allowed to substitute 30 EC of the above-mentioned MSc programme LST by the following Special Programmes.

- Science Education Track of MSc-SEC (info-lst@tudelft.nl)
- Science Communication Track of MSc-SEC (info-lst@tudelft.nl)
- Study abroad (InternationalOffice-TNW@tudelft.nl)
- Choice of courses from the MSc programme Management of Technology (Andrea Witkam, A.A.Witkam@tudelft.nl).

ATTENTION!!!

Students with the exam programme of 2021/22 substitute the elective part (12 EC), the company internship (15 EC), and 3EC of their MSc End Project (MEP) = 30 EC

Students with an exam programme older than 2021, substitute the elective part (12 EC) and the company internship (18 EC) = 30 EC.

These special programmes must be approved by the coordinator of the specialization and the sub-Board of Examiners MSc LST (boardofexaminers-lst-as@tudelft.nl).

NOTE: Some orientations have deadlines for application.

- SCIENCE EDUCATION BASISDEEL: Deadlines for application: 15th November (for semester 2) and 1st of May (for semester 1).

- MANAGEMENT of TECHNOLOGY: Only limited number of places available. Deadlines for application: 15th November (for semester 2) and 1st of May (for semester 1)
Choice for semester 1 or semester 2.

Study Abroad

Introduction 1

Going abroad during your studies is a great way to experience studying and living in a different country, get to know its culture and meeting new people.

The International Office Applied Sciences offers exchange opportunities for both bachelor and master programmes. Other possibilities include summer schools and internships abroad.

Interested? If you want to know more about study abroad, internships and summer schools, please have a look at the topics mentioned under the header Information.

For more detailed information, please see the BRIGHTSPACE pages:

International Office Applied Sciences
Internship Office Applied Sciences

During the academic year students will receive updated information on the possibilities to go abroad.

Year	2022/2023
Organization	Applied Sciences
Education	Master Life Science and Technology

Management of Technology

Contact for Students	info-1st@tudelft.nl
Maximum number of participants	Only limited number of places available. Deadlines for application: 15th November (for semester 2) and 1st of May (for semester 1)

Year	2022/2023
Organization	Applied Sciences
Education	Master Life Science and Technology

1st Semester MoT Modules

MOT1412	Technology Dynamics	5
Module Manager	Dr. C. Werker	
Instructor	Dr. C.C.R. Penna	
Co-responsible for assignments	Dr.ir. U. Pesch	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	In this course we discuss technical innovations and technological development as the result of human choices. Therefore, you learn how to identify the factors that determine these choices, i.e. institutions, behaviour and choices of innovative agents, their relationships, knowledge and collaboration, economic value as well as legitimacy and societal values. Based on that you will figure out how processes of socio-technological change take place as well as study how innovative agents can steer new innovations into a societally responsive direction. You study these issues by using the following theory-based topics: 1) The framework of innovation systems by discussing how innovative agents and institutions drive and shape them on the sectorial/industrial/technological level as well as on the regional, national or global level, in particular by figuring out which role management and policy can play in influencing innovation processes and technological change in different kinds of innovation systems. 2) Methods to integrate societal values into and systems, particularly to find out how to make innovation responsible.	
Study Goals	At the end of the course, students are expected to: 1. analyse innovations as the result of human choices, 2. identify the actions and interactions that are relevant to innovation processes, 3. analyse innovations in a specific technological and geographical context, 4. apply different concepts and methods for specific innovation processes, 5. reflect on the societal implications of innovation and technological change and the challenges that society poses to technology and industry, 6. give recommendations for steering an innovation process into a more societally responsive direction, 7. collect, identify and read relevant literature to do the above mentioned analysis. 8. write up the results of the analyses in a concise and clear way, 9. give feedback to other students.	
Education Method	The course is organized as a series of lectures mostly given in the form of blended learning. Moreover, students can apply the knowledge obtained in the lectures during interactive groups work.	
Assessment	The mark of students is determined by a written exam. Prior to the written exam there are two mock exams: one will cover the first part, the other the second part of the course. Participation in both mock exams as well as correcting the exam of a fellow student both times is mandatory for taking the written exam. Normally the mock exams and written exam take place on-campus. If there are unanticipated circumstances of measures, particularly because of COVID-19, the (mock) exam(s) will take place as a timed open book exam.	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_1FcBgaGLXnjIP0G Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations.	

MOT1421	Economic Foundations	5
Module Manager	Dr. S.T.H. Storm	
Responsible for assignments	Dr. S.T.H. Storm	
Co-responsible for assignments	Dr. C.W.M. Naastepad	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	There is limited participation for students from other Master programs. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation	
Summary	Economics is about how we work, what we produce, how we interact, cooperate and compete, and how we distribute and ultimately use what we have produced. Because the economy is too important to be left to the economists, this course provides an introduction to economics that explains how a market economy works, why it sometimes does not work as well as we would like, how its performance can be improved by regulation and policy, and how over time it changes and develops. The course does not assume that the student has any prior knowledge of economics. The course has two parts: (1) microeconomics, which covers the standard topics of supply and demand, forms of market competition, imperfect competition, efficiency, labour markets and technological progress by firms; and (2) macroeconomics, which examines aggregate demand, growth, welfare, inflation, unemployment and globalization. In both parts, special attention is given to (economic factors explaining) the extraordinary technological dynamism and productivity growth of market economies and to governments role in improving micro- and macroeconomic performance. Microeconomics (2,5 ECTS) focuses on markets, and takes both the firm as the government as principal actors. How are firms' attempts to realize their objectives (in terms of profits, sales or market shares) influenced by demand, technology and market structure? How can public regulation improve market performance which falls below norms considered acceptable? We identify important market failures, due to which markets under-perform, and specifically investigate firm behaviour in (realistic) oligopolistic markets. Macroeconomics (2,5 ECTS) deals with the workings of the economy as a whole, why the market system often underperforms, and how macroeconomic performance can be improved by fiscal and/or monetary policy. Attention will be given to conflicts between achieving different goals of macroeconomic policy at the same time, and to the ways in which globalization affects the capacity of governments to regulate their economies. The answer to this question holds major lessons for public policy.	
Course Contents	Microeconomics: demand and supply; consumer behavior; market competition; perfect competition; monopoly; oligopoly; market failures; externalities; public goods; regulation; efficiency; welfare; production; profit maximization; choice of technique; technological progress; game theory; dominant strategy; labour markets; employment; worker-management relations. Macroeconomics: aggregate demand and aggregate supply; circular flow of incomes; neoclassical economics; Keynesian economics; unemployment; fiscal policy; monetary policy; money creation; crowding out; stabilization policies; labour market regulation; technology policy; foreign direct investment; financial markets.	
Study Goals	At the end of this course, students will be able to: 1. analyze firm behavior in markets of perfect competition, monopoly, and oligopoly as well as the impact of firm behavior on economic welfare. 2. identify and appraise the rationale, scope and limits of public market regulation (in situations of "imperfect markets" and/or "market failures"). 3. derive and document the impact(-s) of regulation on the static efficiency and dynamic efficiency of markets. 4. analyze and weigh up the short-term and long-term impacts of macroeconomic policy on income, employment, productivity and competitiveness. 5. critically examine and reflect on the existing alternative theories and models. 6. understand how money is created and how the financial sector works, and how this affects the real economy.	
Education Method	Lectures. Students are expected to prepare for each class by reading the indicated literature.	
Books	Lecture Notes, Question sets and papers/articles - all made available on Brightspace.	
Assessment	In normal circumstances, the exam is a closed-book (in-person, on campus) multiple-choice examination of 3 hours. In case of unforeseen circumstances or due to measures resulting from COVID-19, the exam will be given in the form of an on-line (remote) open-book & timed multiple-choice examination of 3 hours.	
Special Information	None.	

MOT1442	Social and Scientific Values	5
Module Manager	M.B.O.T. Klenk	
Instructor	M.B.O.T. Klenk	
Instructor	Dr. J.J. Casey	
Responsible for assignments	M.B.O.T. Klenk	
Co-responsible for assignments	Prof.dr. S. Roeser	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	Part I: Scientific Values (Philosophy of Science and Decision Theory) 1. The role of values in science and technology: changing the world vs. describing and understanding the world. The contrast between descriptive claims or arguments vs. normative claims or arguments. 2. The role of reasoning in science. The strength and quality of arguments. The notion of (scientific) proof. The contrast between deductive vs. inductive or ampliative reasoning. 3. The difference between natural and social science. How humans differ from natural things and what this means for the scientific investigation of human behaviour. The contrast between explanatory vs. normative models of action and decision making. 4. The notions of rationality and optimization in action. Different forms of uncertainty and related regimes of decision making. Elementary managerial decision theory. Part II: Social Values (Ethics) 1. Introduction and overview. Getting acquainted with the notions of responsibility and other ethical notions related to managing technology and (individuals working in) the organisations that develop, produce and control technology. Basic meta-ethical questions: is ethics relative and subjective or, alternatively, might there be moral truths or universal moral principles? What are the implications of answers to these questions for management of technology? 2. Introduction into the most important normative ethical theories: utilitarianism and deontology. Exploration how these theories (should) play a role in critical reflection about management of technology. 3. Methods and theories about the right ethical theories to guide action.	
Study Goals	This module explores scientific and moral values in the management of technology. The module presents and discusses science and technology as practices involving a wide variety of judgements that are based on or express values. Values internal to science and technology are discussed in relation to the rationality of belief (e.g. scientific evidence) and action (e.g. decision making). Moral values are discussed by focusing on the ethical and social aspects and problems of technology and of professionals and managers active in the development, production and control of technology, and it explores possibilities for resolving, diminishing, preventing these problems and decision making about them, via for example design for values, responsible innovation and emotional deliberation.	
Education Method	The module is taught in the form of a mixture of (online) lectures and tutorials. Attendance is obligatory. The methodology used is philosophical analysis, reflection and discussion. We will use mostly MS Teams to share information about the course, have online debates, and share educational material.	
Assessment	The assessment method for this course consists of two individual exams and a group-assignment consisting of an essay. Each exam will consist of a multiple choice and an open question part. The final grade for this course is determined according to the following scheme: students receive partial grades for the Scientific Values part of the course and for the Social Values part of the course. The final course grade is the plain average of these two partial grades, provided each is at least 5.76. The partial grade for the Scientific Values part is determined by the grade for a written exam. The partial grade for the Social Values part is determined as the plain average of a grade for a written exam and a grade for a group essay, where both grades need to be at least 5.76. Students do not receive a grade for the course in Osiris until all provisions indicated in the preceding have been met. As long as this is not the case, students can access an overview of their grades so far in Brightspace. Sufficient grades received for assessment components remain valid until a sufficient grade for the course has been reached. If unforeseen circumstances occur, such as COVID-19 related measures that prohibit exam-taking on campus, we will deviate from the above determined assessment method as follows: The individual exam(s) affected by the circumstances will be held online as a Möbius open book exam.	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_3RkkO6GetqrStb8 Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations	
Tags	Ethics Philosophy	

MOT1461	Financial Management	5
Module Manager	Dr.ir. Z. Roosenboom-Kwee	
Instructor	A.A. Ralcheva	
Responsible for assignments	Dr.ir. Z. Roosenboom-Kwee	
Responsible for assignments	Dr.ir. Z. Roosenboom-Kwee	
Co-responsible for assignments	Dr. S.T.H. Storm	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Expected prior knowledge	None	
Summary	This course is about how corporations make financial decisions. It is designed to equip technical/engineering students with fundamental financial management skills.	
Course Contents	The course covers topics such as financial statement analysis, types of financing and investment decisions, and financial consequences of those decisions. While investment decisions involve spending money, financing decisions involve raising it. Important to highlight is that corporations finance their investments by borrowing, by retaining and reinvesting cash flow, and by selling additional shares of stock to the corporations shareholders. Hence, there will be among others three broad financial questions discussed in the lectures. First, what investments should the corporation make? Second, how should it pay for those investments? Third, what plans can the firm make to ensure that it stays financially strong and healthy?	
Study Goals	At the end of the course, students are expected to acquire the ability to: evaluate financial performance of companies through financial statement analysis evaluate investment opportunities using investment appraisal techniques such as internal rate of return (IRR), net present value (NPV), payback period, profitability index understand and discuss different types of long-term (debt and equity) financing such as venture capital, initial public offering, corporate debt) financial instruments and markets perform valuation of bonds and stocks analyze choices of capital structure and portfolio investment evaluate investment and capital structure decision risks (e.g. financial distress) understand and analyse dividend payout policy	
Education Method	The lectures and workshops are on campus. However, depending on the COVID-19 measures, the lectures and workshops may be given through a hybrid form or through online and remote learning.	
Literature and Study Materials	Lecture slides and textbook	
Books	Jonathan Berk and Peter DeMarzo (2020). Corporate Finance: The Core. Fifth Edition, Pearson. ISBN: 9781292304151. The fourth edition is also acceptable.	
Assessment	The course assessment is a written exam through Digital exam on campus. The exam is open questions and closed book. Only a calculator and a pen are allowed. Dictionary is not allowed. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows: Digital Exam (to be announced)	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_bdQf6ZAuEILSEjI Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations	

MOT1524	Leadership and Technology Management	5
Module Manager	Dr. R.M. Verburg	
Responsible for assignments	Dr. R.M. Verburg	
Co-responsible for assignments	Dr. L. Rook	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	Leadership and management in the context of high technology firms and digital business are the key theme of this course. The core of successful management in technology-based environments is the recognition of the value of employees for organizational success. Selection, development and performance management of employees are core activities of personnel management. Such practices are no longer seen as tasks that are the responsibility of a separate personnel department, but they become part of top managements personnel strategy. Line managers are increasingly expected to play an active role in implementing management policies and practices. Leadership appears to be a critical determinant of organizational effectiveness whether we are discussing a football team, an army, a street gang or a multinational corporation. Leadership is the subject of intense academic research and debate and journalistic commentary. In this course we will highlight the critical roles of leadership on different levels for the success of technology-based companies.	
Study Goals	At the end of this course, students are expected to competently: - analyze the nature of leadership management within advance technology organizations - provide arguments how leaders contributes to organizational performance - recognize knowledge processes in organizations - recognize management practices and analyze the success of these within different organizations - show the need for alignment between management practices - show the need for alignment between management practices and business strategy - recognize best practices with regard to the selection and performance management of employees - qualify scientific research in the area of Leadership and Organizational Behavior - recognize the current and future challenges of technology firms in leading and managing people	
Education Method	The course consists of weekly lectures. The lectures are dynamic and participative including: - Theory presentations and interactive lectures by the staff (in class and online) - Presentations by a guest speaker from practice - Presentations by students who work together in expertise groups - Case write-ups	
Assessment	Individual final examination The final exam will be a closed book exam on the content of this course. Hence, students who take the course seriously and structurally prepare the literature and assignments will be very likely to pass the exam. The exam will consist of Open Questions. A minimum grade of 5.75 in the exam is necessary for a passing this course. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows: Take-home assignment (open book), i.e. an Individual assignment based on open-ended question(s), typically to be delivered within period of a few days	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_8jZbDsClxC1FHBV Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations	

MOT2313	Research Methods	5
Module Manager	Dr.ir. G.A. de Reuver	
Instructor	Dr. L. Rook	
Co-responsible for assignments	Dr. L. Rook	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	Student learn how to design and execute research, as well as how to interpret and critically assess the outcomes of research. Both quantitative and qualitative research methods are taught. To do so, students learn the fundamentals of research design as well as quantitative and qualitative research methods. This course is designed to help students develop and execute research steps from the initial research problem until drawing conclusions. These steps include conceptualization of research ideas/problems, development of a relevant theoretical framework and related hypotheses, defining and operationalizing core concepts, and developing an appropriate research design. Students learn how to scrutinize the findings and draw implications drawn from statistical analysis and qualitative analysis of data. Topics that will be covered among others: - define research problem, research objectives and questions - conduct research process of scientific investigation - develop a conceptual model - formulation of hypotheses - operationalize and develop measurement of concepts - develop research design, as well as make motivated decisions on data collection and data analysis methods - evaluate different types of quantitative techniques (hypothesis testing, non-parametrics, regression analysis, multivariate analyses) - interpret and critically reflect upon findings from statistical analysis - qualitative data analysis, based on coding principles - interpretation and presentation of results - reflection on reliability and validity of research results - research ethics	
Study Goals	1. Create a research design for a research problem relevant to the context of MOT To do so, sub-goals are to: 1a. Understand core concepts of research design (i.e. theory, population, sample, hypotheses, research strategy, operationalization, data collection) 1b. Apply these core concepts to design a research project (i.e. from a broad topic towards research questions, conceptualization, research strategy, research design, data collection approach, sampling, operationalization, and selection of analysis methods) 2. Critically evaluate outcomes of qualitative and quantitative research in the MOT context To do so, sub-goals are to: 2a. Interpret statistical analysis findings 2b. Scrutinize / Reflect upon findings / conclusions from both qualitative and quantitative research (i.e. validity, reliability, traceability, ethics) 2c. Suggest alternative / future research directions	
Education Method	Course contains three main parts: 1. Research design 2.1. Qualitative research methods 2.2. Quantitative research methods Part 1 and 2.1 consist of: - Lectures, which are a mix of instruction and formative assignments. - Summative assignments (weekly, group of 2 students) in which a research design is created Part 2.2 consists of: - Lectures and practice	
Literature and Study Materials	To be announced via BrightSpace. At least one textbook on research methods for business.	
Assessment	Summative assignments (15% = 0.75 EC): Group of 2 students, weekly assignment, week 1-5 of the course. The summative assignments cover part 1 and 2.1 of the course. Digital exam (85% = 4.25 EC; at least 5.0 to pass the course). The exam covers part 1, 2.1 and 2.2 of the course. The exam contains open questions and is closed book. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed exam will be modified as follows : Timed digital (Möbius) open book exam	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_3ObBU6LgvAgQ0Hs Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations	

Year	2022/2023
Organization	Applied Sciences
Education	Master Life Science and Technology

2nd Semester MoT Modules

Introduction 1	- MANAGEMENT of TECHNOLOGY: Only limited number of places available. Deadlines for application: 15th November (for semester 2) and 1st of May (for semester 1) Choice for semester 1 or semester 2. Take a look at the 1st en 2nd semester courses of MoT in the digital Studyguide.tudelft.nl and search for Faculty TPM - Management of Technology.
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MOT1003	Integration Moment	5
Module Manager	Dr. R.M. Verburg	
Instructor	Prof.dr.ir. A. Verbraeck	
Instructor	Dr.ir. G.A. de Reuver	
Instructor	Dr. N. Pachos-Fokialis	
Instructor	H.A. Ofe	
Instructor	Dr. S. Hinrichs-Krapels	
Responsible for assignments	Dr. R.M. Verburg	
Co-responsible for assignments	Prof.dr.ir. A. Verbraeck	
Contact Hours / Week x/x/x/x	0/0/0/9	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	Please, be aware that students, who are not enrolled in the MOT program, cannot take this course as an elective. This course constitutes an integration of the Management of Technology master, which requires knowledge of the core modules of the program. Therefore, the course is not an elective but for MOT students and for a selected group of students who are enrolled in the MOT orientation program only.	
Course Contents	The Master program in Management of Technology (MOT) educates students with a bachelor degree in engineering as technology managers, innovation managers, analysts of technological markets (either as scientists or consultants), and entrepreneurs in technology-based, internationally-oriented competitive business environments. The ultimate objective of the program is to improve the quality of technology and innovation management in the different engineering mono-disciplines in practice. Students contribute to scientific work in the area of MOT and have the opportunity to apply their knowledge in real-world settings, such as advanced technology organizations, laboratories, and high-tech business ventures. The program deliberately aims at an international and diverse group of students. Students of MOT are all rooted in different engineering mono-disciplines and work together in order to combine insights from the different engineering disciplines and to apply the different aspects of technology and innovation management. In this particular course, we aim to integrate knowledge and skills as put forward in the first year of the MOT program by applying this knowledge and skills through a consultancy assignment for MOT related business cases of organizations and companies. You will work as a technology management consultancy firm for a client with the aim to reflect on knowledge and skills needed to work in a typical MOT environment.	
Study Goals	At the end of this course, students are expected to: At the end of this course, students are expected to: 1. Apply MOT concepts from the first-year program on a case study. 2. Show a thorough understanding of the (inter-) relationships between the different parts of the MOT program. 3. Write a well-argued consultancy report and present your advice to a client.	
Education Method	Class members are divided into teams of three and assigned the task of doing a consultancy project. Each team will collaborate with three other teams on a business case for an actual client. Participants are supported by different MOT coaches during the progress of the assignment. The aim is to provide a solution to a current business problem in a coherent consultancy report in which the following four aspects as underlined in the MOT program are integrated A. Organizational angle (see e.g. MOT1524, MOT1531), B. Commercial angle (see e.g. MOT1434, MOT2421), C. Financial (investment) angle (see e.g. MOT1421, MOT1461), D. Safety, security, responsibility angle	
Assessment	Each of the partial reports will be graded on a scale from 1-10. The total report will be graded from a consistency standpoint as well, which can influence the mark of the partial report for a group. The grade for the final report will be determined by your case coach in collaboration with a specific content coach for technology, innovation, finance and security/safety on the basis of a detailed grading grid. The grid includes the following criteria: 1. The use of MOT concepts and theories(1-10) 2. Understanding of the core issue/topic(1-10) 3. Advice to the client (1-10) 4. Argumentation, and quality of the report(1-10) 5. The contribution of your particular group topic(1-10) The five criteria will weigh equally. You need a minimum of a 6,0 in total to pass the course. As the grading is project- based, the re-take opportunities are limited to re-writing (parts of) the report in the case of a grade lower than 6,0.	

MOT1435	Technology, Strategy and Entrepreneurship	5
Module Manager	Dr. G. van de Kaa	
Instructor	Dr.ing. V.E. Scholten	
Instructor	Prof.dr. C.P. van Beers	
Responsible for assignments	Dr. G. van de Kaa	
Co-responsible for assignments	Prof.dr. C.P. van Beers	
Contact Hours / Week x/x/x/x	0/0/5/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Summary	The course Technology, Strategy and Entrepreneurship focuses on formulating and implementing technology strategy for large firms and entrepreneurs. However, before a manager can formulate a successful strategy, it is needed to understand the specifics of the external economic and societal environment in which the strategy will be implemented. Throughout the course, students will be acquainted with a variety of academic perspectives which are utilized by strategy and entrepreneurship researchers.	
Course Contents	Greater product and technology complexity and increased internationalization of innovation activities have increased costs and risks for innovators such that the boundaries of the firms have changed and are changing. Specifically, we explain the typical patterns of technological innovation in order to provide a foundation for the formulation of a technology strategy. Furthermore, we focus on identifying the reasons why most industries adopt a dominant design and why a particular firm's technology is adopted as the dominant design. We will examine the factors that affect the outcome of standards battles. These include e.g. timing of entry strategies, collaboration strategies, and appropriability strategies. We will provide a basic background of entrepreneurship. Specifically, we will pay attention to networking strategies for new firms (entrepreneurs). How do these firms attempt to achieve success with their new technology, in a new market? As the boundaries of the firms are getting more transparent due to greater complexity of technical projects, it is required to decide whether to collaborate with external partners, and with whom. This can be a good option for entrepreneurial firms as they lack the necessary complementary assets to successfully commercialize a new technology. We will explore collaboration strategies for entrepreneurial firms. Technological innovation is a much more uncertain activity for a firm than other activities. Strategic thinking and formulation require knowledge on the boundaries of the firm and the markets in which the firm operates, its value chain and its core competencies. Also, strategy formulation aims at controlling the external environment of the firm, which is characterized by uncertainties. A distinction can be made in changes in the external environment in time or by location. General Purpose Technologies affect strategy formulation in time and frugal or inclusive innovations by location. After formulating the sources and innovation strategies we will pay attention to the resources firms have for innovation processes and specific valuation techniques, like real options, conjoint analysis and data envelopment analysis.	
Study Goals	After the course students are able to - understand the theoretical background of technology and innovation strategy. - position a variety of strategy topics thereby taking into account sectoral and country context differences. - understand the theoretical background of entrepreneurship. - apply key technology strategy models in different sectoral and country contexts. - understand the relations between those models. - analyse the industry dynamics of technological innovation. - analyse companies technology strategy. - analyse and explain a company's technological innovation strategy vis-a-vis its strategic or entrepreneurial posture - analyse and explain the organization and structure of new product development teams based on the firm's technological innovation strategy	
Education Method	This is a blended course and the course includes regular sessions, games, and teaching cases. Each week will start with a spark which will be communicated through Brightspace. On the first day the teacher will discuss the spark with you. On the fourth day of the week, a regular interactive lecture is planned. On the fifth and final day of the week quizzes are available. Each regular session will consist of lectures. In the teaching case sessions students discuss questions posed in assigned teaching cases that relate to the topics studied in the regular sessions. For the regular sessions students study the literature and for the teaching case sessions students prepare the teaching cases. In the teaching games students apply the theory by implementing and evaluating various strategies in a real life situation.	
Literature and Study Materials	Book: Schilling, M. A. 2019. Strategic Management of Technological Innovation. New York, USA, 6th edition Journal articles: Will be posted in Brightspace.	
Assessment	Slides presented in class: Will be posted in Brightspace. The final grade for this course is determined as follows: 1. Individual exam on book, literature and lectures (70%) 2. quizzes (30%) o In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows: the individual exam will be replaced with a timed take-home exam (open book). This will be a remote examination.	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_3EpjZgyQcSYv7Fk Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations	

MOT1452	Inter- and intra-organisational decision making	5
Module Manager	J. Lieu	
Instructor	Prof.mr.dr. J.A. de Bruijn	
Instructor	Dr. W.W. Veeneman	
Instructor	J. Lieu	
Instructor	Dr. L.D. Virla Alvarado	
Responsible for assignments	Dr. W.W. Veeneman	
Co-responsible for assignments	J. Lieu	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	All organizations whether large or small, public or private, old or new make decisions on a regular basis. One might assume that these decisions taken inside and between organizations are well-planned and based on sufficient information. But in reality, information is often incomplete and contested among the actors making a decision. These actors may also have competing interests, for instance governments that favor different policies, or companies that pursue their individual profits. As a result, complex decisions often appear chaotic and ill-structured. The primary purpose of this course is to help understand why it is thus impractical to solve multi-actor decision problems using a purely analytical approach; Why carefully, pre-conceived plans can backfire when hierarchically imposed, and force even ministers and CEOs to the negotiation table; And why decision-making in networks can be characterized as a permanent struggle and exercise of power among interdependent actors. The course introduces students to theories and models which help explain how decisions are made in practice, i.e. what decision-makers really do, and why they do it that way. It acquaints students with networks and strategies that can be used to manage actors and interests. During lectures, students apply these theories to real-worlds examples through group discussions. In parallel, they analyze a case study in depth, reconstructing and evaluating the decision-making process, and writing a paper about it. By the end of the course, students will learn to design processes and strategies that enable change in multi-actor settings.	
Study Goals	At the end of this course, students will be able to achieve the following learning objectives: 1: Explain key concepts, perspectives, as well as negotiation and framing strategies on decision-making within multiple-actor networks; 2: Explain individual and collective decision biases in different types of organisations 3: Evaluate the decision process surrounding a complex real-world problem 4: Analyse a decision concerning a recent real-world wicked problem 5: Apply negotiation and framing strategies used in multi-actor decisions	
Education Method	This course consists of lectures, tutorials, self-study, and assignments. During lectures, we will synthesize the required readings. News events and large projects are often used as examples. (Students are expected to have read the relevant literature beforehand). During tutorials, students can discuss their papers as they progress with their case analysis. Interim assignments help students along the way.	
Literature and Study Materials	The required reading for the course includes (chapters of) the following books: 1. "Thinking, Fast and Slow", by Daniel Kahneman, 2013. 2. Hans de Bruijn et al., Management in Networks, Chapter 4: A process, not a project. Routledge, London. DOI: 10.4324/9781315453019 In addition, a number of book chapters and papers are uploaded/linked to on Brightspace. Some of them are mandatory reading; others give additional context or examples.	
Assessment	The final grade for this course is determined as follows: 1. Individual Assignment (35%): critical reflection of theoretical knowledge and reflection of personal learning process 2. Participation (15%): assess theoretical knowledge and motivate active participation and discussions during lectures. 3. Group Paper (50%): assess the ability to apply theory to a real case and write a well-argued paper in a multi-actor team setting. To pass, students need a minimum of 5.8 on both the individual assessment and group paper. Group papers are only graded if the mandatory assignments have been submitted on-time.	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_2rY1TinxlrEIOFY Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations	

MOT1531	Digital Business Process Management	5
Module Manager	Prof.dr.ir. M.F.W.H.A. Janssen	
Instructor	Dr. J. Ubacht	
Instructor	Dr. Y. Ding	
Responsible for assignments	Prof.dr.ir. M.F.W.H.A. Janssen	
Co-responsible for assignments	Dr. J. Ubacht	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	This course has a maximum capacity of 200 students. There is limited participation for students from other Master programs. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	Business processes are at the core of high-tech organizations and consist of a large number of activities aimed at creating value for customers. These processes can be executed by humans or by digital technology. In the digital age business processes are supported by, enabled and changed by technologies like the Internet of Things, Artificial Intelligence, Robotics and so on. Business processes are situated at the point where the business side meets the IT side and where technology should fulfill customer needs. If business processes fail or do not adapt to the needs of customers the whole business might fail. Business Process Management (BPM) is necessary for ensuring that processes are operating in concert and that these processes are adaptable to changes in the overall environment of an organization. In this course the relationship between strategy, technology and business processes will be illustrated, explained and discussed. Methods and tools for business process analysis and improvement will be presented, including a number of statistical and ICT-architecture methods. New solutions will be architected for high-tech organizations. The extent to which the nature of business processes is affected by contemporary technology constitutes the core of this course. In particular to analyze technologies and their impact on business process management and to architect new solutions in the organizational context of the firm. Topics - Aligning strategy and processes, value perspective, supply chains - Workflow, resources, BPM elements and control - Business process modeling (concepts, BPMN) + practical work - Process measurement, Balanced Score Card, Activity-based Costing (ABC) + practical work - Business process improvement strategies and approaches, path dependencies, capabilities, resource-based view, business process maturity - Improvement methodologies (LSS, lean, six sigma, theory of constraints, technology ..) - Statistical analyses and improvement + practical work - Compliance by design, build-in-controls in processes and technology - Business process automation and technology (BPM, ERP, workflow, SOA,...) - Business processes automation (case-based, process -based, use-based) - Agility and adaptability of BPM systems and the users, business process and agile enterprises - Business and knowledge rules - Internet of Things, data-driven organizations	
Study Goals	The overall aim of the course is to learn how to improve the relationship between business processes, strategy and technology for realizing organizational strategies. The focus is on realizing business processes enabled by digital technology for high tech organizations. In particular the objectives are - To analyze the relationship between business processes, strategy and technology - to understand new technologies and their impact on business processes - to be able to select and use methods and tools for improving business processes - to analyse and provide improvement suggestions for business processes and technology	
Education Method	-lectures -practical work -assignment	
Computer Use	Planning tool. There are suitable planning tools like Trello and Asana BPMN modelling tool. There are several modelling tools which are appropriated like Eunomia process builder (http://www.eunomia-process.com/), Bizagi (http://www.bizagi.com/) or LucidChart (https://www.lucidchart.com/) Activity Based Costing (ABS) calculations and simulations using a spreadsheets, like MS Excel.	
Literature and Study Materials	Parts of the book N. Bharosa, R. Van Wijk, N. De Winne & M. Janssen (2015). Challenging the Chain. Governing the Automated Exchange and Processing of Business Information. IOS Press http://www.iospress.nl/book/challenging-the-chain/ - open access ebook List of articles (containing classical papers and showing the state-of-the-art) Reader (containing an overview of main elements and explaining some methods) Exercises + answers (for practicing)	
Assessment	Students will collaborate in groups to make an assignment which should contain 1) the analysis 2) improvement and digitalization of a business process and 3) implementation aspects. Students are free to select a process that is critical or a company and can be improved by ICT. The assignment consists of three parts 1. Analysis: Students select a company, analyse their strategy and select a critical business process. The existing process should be modeled (in BPMN) and the resulting product, stakeholders role,s customer needs and technology used should be analysed. 2. Improvement: Based on the improvement methods and instruments a proposal for business process improvements should be made. 3. Realization: how the proposed business process should be realized by means of digital technology. The assignment should be handed in as a report. The report should adhere to the APA reference standards. The students are expected to prepare two presentations and some of the groups will be asked to present these during the lectures. Based on these presentations feedback will be given, which are relevant for all groups. Presentations will be assessed using a pass or fail. The report will be graded; this grade will reflect the final grade.	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. Other students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_787nY10vO4JW7zg Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations You can apply until January 22.	

Elective	Yes
Tags	Economics Information & Communication Modelling Technology

MOT1534	High-tech Marketing	5
Module Manager	Dr. L. Rook	
Instructor	Dr. L. Rook	
Co-responsible for assignments	Dr. J.R. Ort	
Contact Hours / Week x/x/x/x	0/0/2x2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	Companies in high-tech markets face constant uncertainties caused by market dynamics, technological progress and volatile customer preferences. In order to survive under such highly dynamic circumstances, companies must keep on innovating, and carefully select -- or modify -- appropriate high-tech marketing strategies. This course aims at familiarizing students with several theoretical and methodological approaches to high-tech marketing, and will also capture recent developments in marketing research, such as neuromarketing, and Web-based (2.0) and Mobile-based (3.0) technologies that render high-tech marketing prone to (online and offline) word-of-mouth dynamics. These issues will be discussed under reference to best practices (from various high-tech companies) that serve to illustrate ways in which to deal with high-tech marketing strategy problems.	
Study Goals	At the end of the course, students are expected to have gained knowledge on principles of high-tech marketing and will be able to recognize, analyze, reflect upon, and creatively respond to, high-tech marketing problems and challenges with regard to: - Marketing strategies and plans to deal with uncertainty in high-tech environments; - Customer segmentation, targeting, and positioning for business-to-business (B2B) and business-to-consumer (B2C) markets; - Product and service bundles development, (offline and online) word-of-mouth, advertising and promotion, incl. modern online tools; - Some marketing research techniques (various methodologies) aimed at analyzing customer preferences for regular as well as persuasive (Web 2.0/3.0) marketing strategic communications and technologies (recommender systems).	
Education Method	The course will consist of a series of lectures based on: - Theory discussions and interactive lectures based on the course materials; - Guest lectures from experts from academia and practice; - Analysis and evaluation in small groups of existing corporate marketing strategy problems in a self-chosen set of case companies. Analysis will be based on strategic assessment of the business case, and should be grounded in a customized marketing plan, and result in a creative proposal for a business-to-business (B2B) or business-to-consumer (B2C) marketing campaign. All these components in the mix should lead to specific recommendations on how the case company needs to cope with (some of) the most pressing high-tech marketing challenges. Individual academic reflections on the groupwork are required.	
Literature and Study Materials	The materials for this course are: - A course book (see below). - Academic articles from leading journals in Marketing Research (a.o., Marketing Science, J. of Marketing, J. of Marketing Research, J. of Consumer Research) and in the Automation, Computing Machinery (ACM) research community to be accessed via the TU Library online repositories.	
Books	t.b.a.	
Assessment	Students will be graded on the quality and rigor of a marketing research project to be executed in group setting, complemented with individual theoretical reflections upon the group assignment. Specifically: - A substantial group paper will (a) critically review the relevant literature on high-tech marketing to a relevant business case. At the heart of the group paper lies (b) empirical (qualitative and/or quantitative) analysis of a specific marketing problem, and (c) the adequate discussion thereof; analyses will inform (d) the development of a creative marketing campaign intended to solve the business challenge identified. - Each group paper must be supplemented with contributions (theoretical reflections) from each of the individual group members. - The research project may include peer review so as to determine the weight of individual contributions to the group work. - The assessment of the marketing research project will be based on a detailed grading rubric shared in advance. - A minimum final grade of 5.75 is a requirement for passing the course.	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_24ybEwlWiOwgnFs You can sign up until Sunday, January 22, 2023. Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations	

MOT2421	Emerging and Breakthrough Technologies	5
Module Manager	Dr. J.R. Ortt	
Responsible for assignments	Dr. J.R. Ortt	
Co-responsible for assignments	Dr. L. Rook	
Contact Hours / Week x/x/x/x	0/0/0/8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	The course focuses on innovation processes in a high-tech environment. Firstly, innovation processes are studied on the micro-level, distinguishing stage-gate, agile and hybrid project approaches to innovation. Secondly, the course will discuss the development and diffusion of breakthrough technologies on the macro-level. The pattern, of the development and diffusion of such technologies, the actors and factors that explain the evolution of the pattern, and strategies that companies can adopt during the pattern are discussed. Thirdly, the course will shortly discuss how disciplines (defined as combination of related technologies) emerge and evolve. The three levels of analysis (project, pattern, discipline) together form a comprehensive perspective on innovation processes.	
Study Goals	Describe innovation processes on three levels: (1) project level, (2) level of the pattern of development and diffusion of technologies and (3) discipline level. Distinguish alternative theoretical perspectives on each level and the assumptions of these perspective. Be able to explore and analyse an innovation opportunity or problem, choose the level of analysis and apply the perspective on that level that fits the context of the opportunity or problem.	
Education Method	lectures and assignment. Normally lectures are face-to-face. If needed, due to circumstances, lectures will be on-line.	
Assessment	Exam and assignment: In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows: 1. The exam will administered on-line using maple-ta or a similar program. 2. The assessment will remain the same.	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_6mrjadvsL8ud8Eu Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations	

Year	2022/2023
Organization	Applied Sciences
Education	Master Life Science and Technology

Honours Programme 2022

Introduction 1

The Honours Programme consists of 20 EC on top of the regular master programme of 120EC. It should be finished within the standard two years.

The Honours Programme consists of a collective part (5 EC) and a individual part (15 EC)

You are eligible for an honours programme when your study results are on track; you must have a GPA of 7.5 or higher. In the month of February of the academic year, students who fulfil this requirements, are invited to join an informative meeting on the Honours Programme

The Honours Programme must be approved by the programme director, Prof dr Ulf Hanefeld, U.Hanefeld@tudelft.nl.

Year	2022/2023
Organization	Applied Sciences
Education	Master Life Science and Technology

Collective Part 2022

UD2010	Critical Reflection on Technology	5
Responsible Instructor	Dr. P.E. Vermaas	
Course Coordinator	Dr. P.E. Vermaas	
Contact Hours / Week x/x/x/x	X/x/0/x	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The course is organised in three parts. Part 1: The course starts with 5 class meetings in which we discuss texts about technologies from the social sciences and philosophy. These are provocative texts: they are engaging and (should) make you think. They have to be read in advance. Topics of the texts include technological surveillance, climate changes, AI and human enhancement. The goal is to get acquainted with different styles of argumentation, and with discussing subject matters with audiences. Part 2: During the course each student will be individually supervised by a tutor to develop a pitch presentation in which the student presents and argues for a challenging position on a societal issue related to technology. The choice of this position is in principle free but is typically focussed on one's individual honours program. A pitch consists of (i) a 900 to 1100 words paper (excluding references) in which the position and argumentation of the pitch is described in detail, and (ii) a five minutes presentation in class. Important in the assessment of the pitch is the extent to which the pitch can convince its intended audience. Part 3: At the end of the course two meetings are organised in which students present their pitches and defend them.	
Study Goals	The ability to form an independent, well-argued position with regards to ethical or societal issues concerning technology in general or the professional practice of engineers more in particular. The ability to discuss ethical or societal issues concerning technology in a public setting.	
Education Method	Part 1: Workgroup discussions with brief lectures; Part 2: Individual supervision; Part 3: Presentation + defense.	
Assessment	Participation in all class meetings and in one presentation meeting is obligatory; in exceptional cases a student is allowed to skip at most one class meeting during the whole course. The assessment will be based on the pitch and on the participation in the discussions during the class meetings. The final result will be either pass or not-pass.	
Remarks	This course is run two times: in Q2 and in Q4 Please note: this course is open to Honours Programme Master (HPM) students only!	

Year	2022/2023
Organization	Applied Sciences
Education	Master Life Science and Technology

Individual Part 2022

Year	2022/2023
Organization	Applied Sciences
Education	Master Life Science and Technology

1. Company Oriented HPM 2022

AS1011HPM	Applied Sciences Company Project	12
Responsible Instructor	Prof.dr. J.J.C. Geerlings	
Contact Hours / Week x/x/x/x	Different	
Education Period	Different, to be announced	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Required for	Company Oriented Honours Programme Master of the Faculty of Applied Sciences.	
Course Contents	Master honours students from the Faculty of Applied Sciences work in multidisciplinary teams of 3-6 persons on a company project. The project deals with actual question of technical or more future scenario oriented problems. During the project, the teams will use expertise from their own fields of studies, to deliver a final advice or product (depending on the assignment). A final report and presentation will be given to the company after the assignment is finished.	
Study Goals	The work done, and results delivered demonstrate that as a student, you are able to: - Systematically analyse the company's problem statement, based on initially (possibly unclear) goals and deliverables; - Apply academic qualities (theorizing, argumentation and reasoning) to proposing solutions to a real company problem; - Constructively and collaboratively discuss your work within a multidisciplinary team (with people from different educational and cultural backgrounds) - Systematically evaluate the possible company problem solutions to select the best product / advice with regard to the problem statement; - Creatively and credibly assemble arguments to design a project advice report / new product recommendations; - Write a professional joint project report for a company with a multidisciplinary team of students; - Presenting the results of your project to a professional audience.	
Education Method	Multidisciplinary teams work on a real-life company problem, under supervision from the TU Delft and the company.	
Course Relations	A series of workshops (Honours Classes of Applied Sciences) will run parallel to this course, covering topics ranging from marketing strategies and communication, to negotiation and advising. The exact content of these workshops will depend on the input that the students need in their Company Projects.	
Assessment	For the Company Project, students will be assessed on the content and quality of the final group report; the quality of the groups presentation to the company and the assessment by the company.	
Enrolment / Application	This course is open to students that participate in the Company Oriented track of the Honours Programme Master of the Faculty of Applied Sciences.	

AS1021HPM	Applied Sciences Honours Classes	3
Responsible Instructor	Ir. S.M. Flipse	
Contact Hours / Week x/x/x/x	Different	
Education Period	Different, to be announced	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Required for	Company Oriented MSc Honours Programme of the Faculty of Applied Sciences.	
Course Contents	The content of this series of Honours Classes supports the Company Project of the Master honours students of the Faculty of Applied Sciences. In these Honours Classes, students reflect on the problems they are solving during their Company Project. The content of these Honours Classes relate to e.g. corporate strategy, image and identity, marketing approaches, teamwork and collaboration, advertising and promotion, reasoning and argumentation, communication and negotiation, and credibility. Depending on the needs of the students and their assignments, the theme order and priorities may vary. These Honours Classes depart from theoretical backgrounds on these themes, which are made practical during five consecutive interactive sessions.	
Study Goals	The work done, and results delivered demonstrate that as a student, you are able to explain and analyse the Company Project from a theoretical background on corporate strategy, image and identity, marketing approaches, teamwork and collaboration, advertising and promotion, reasoning and argumentation, communication and negotiation, and credibility.	
Education Method	Interdisciplinary Honours Classes, linking science communication related theories and approaches to a real-life company problem during five consecutive interactive sessions.	
Course Relations	This course runs parallel to the Company Project of the Faculty of Applied Sciences MSc Honours Programme. The students use insights from the Honours Classes in their company projects.	
Assessment	For each of the themes covered in the various Honours Classes, students write an individual reflection on how they used this theme in their Company Project (max 1A4 per theme). Scoring rubrics will be distributed via Brightspace.	
Enrolment / Application	This course is open to students that participate in the Company Oriented track of the Honours Programme Master of the Faculty of Applied Sciences.	

Year	2022/2023
Organization	Applied Sciences
Education	Master Life Science and Technology

2. Consistent package of challenging courses 2022

Contact for students	info-1st@tudelft.nl
Director of Education	prof. dr Ulf Hanefeld - U.Hanefeld@tudelft.nl
Prerequisites	Master students with an excellent performance during their first semester, i.e. a GPA of 7,5 or higher. The application procedure includes a motivation letter and a short interview. After admission to the programme you will be registered for the Honours Programme Master.
Introduction 1	The Honours Programme must be approved by the Programme Director, prof.dr Ulf Hanefeld
Introduction 2	Coherent set of challenging course modules of at least 15ec around a theme with motivation. Either a focus on broadening, or a focus on deepening. It's NOT a set of some additional regular electives from the MSc programme that can't be distinguished from your other electives. The set of courses must be approved by the programme director. An Honours Programme is not just a package of courses on top of your Master programme; it has to be something special and challenging for you.

Year	2022/2023
Organization	Applied Sciences
Education	Master Life Science and Technology

3. Highlights in Biotechnology 2022

LM3351	Topics in Biotechnology theory	0
Responsible Instructor	C.E. Paul	
Responsible Instructor	Prof.dr. U. Hanefeld	
Contact Hours / Week x/x/x/x	0/0/x/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	BSc degree LST or similar. The course focusses on students joining in February rather than at the general starting time in September, at exchange students and at honours students.	
Course Contents	THE COURSE LOAD VARIES BETWEEN 3 - 5 EC The course will focus on a current high profile topic in the life sciences & technology. This topic will alter every second year and will be chosen by the teacher of the course. Details will be given by the teachers. The topic will be taught in lectures and self-study and will be finally evaluated via an essay of adaptive length to suit the desired number of ECTS. 2022/2023: Biocatalytic retrosynthesis. The course will focus on the retrosynthetic approach using enzymes. Recent development of biocatalysts that can be applied in organic synthesis allows to rethink the way in which target molecules might be constructed. Principles of biocatalytic retrosynthesis with consideration for strategic disconnections, functional group interconversions, and structure-guided analyses will be studied. A work plan will be made on how to disconnect a target molecule into identifiable building blocks that can be produced by biocatalysts.	
Study Goals	Explore an unknown current research topic via literature, guided reading and discussion After the course the student will have developed in depth knowledge on the topic of the course and have mastered the skills of writing an essay. This includes: - Understanding why the topic is considered high profile - Precise and concise formulation of the topic and its relevance to Life Science & Technology and to society in general - Explore an unknown current research topic via literature, guided reading and discussion	
Education Method	Lectures, Self-study, research plan	
Literature and Study Materials	References made available on brightspace	
Assessment	Essay of adaptive length to suit the desired number of ECTS (THE COURSE LOAD VARIES BETWEEN 3 - 5 EC)	
Special Information	The course load varies between 3 - 5 EC, depending on the 'Topic of the year' and/or the desired length of the essay.	

LM3356	Topics in Biotechnology practical	0
Responsible Instructor	C.E. Paul	
Responsible Instructor	Prof.dr. U. Hanefeld	
Contact Hours / Week x/x/x/x	0/0/x/x	
Education Period	3 4	
Start Education	3 4	
Exam Period	Exam by appointment	
Course Language	English	
Required for	The course focusses on students joining in February rather than at the general starting time in September, at exchange students and at honours students.	
Expected prior knowledge	BSc degree LST or similar. Topics in Life Science & Technology theory.	
Course Contents	THE COURSE LOAD VARIES BETWEEN 10 - 12 EC, depending on the 'Topic of the year' and/or the desired length of the essay. The course will focus on a current high profile topic in the life sciences & technology. This topic will alter every second year and will be chosen by the teacher of the course. Details will be given by the teachers. The topic will be taught in laboratory work and self-study and will be finally evaluated via the BEP rubric to suit the desired number of ECTS. This year 2022/2023 the biocatalytic retrosynthesis approach is the target of the course. In the theory part of the course the students will have developed a plan on how to disconnect a target molecule into identifiable building blocks that can be produced by biocatalysts. In the second part this theoretical exercise will be put into practice.	
Study Goals	After the course the student will have developed and applied in depth knowledge on the topic of the course and have performed an independent piece of practical research. This includes: - Understanding why the topic is considered high profile - Precise and concise formulation of the topic and its relevance to Life Science & Technology and to society in general - Explore an unknown current research topic via literature, guided reading and discussion - Formulating a research goal based on the above - Formulate a research plan on how to perform this research and execute it - Write a report on the achieved results	
Education Method	Practical work in the laboratory	
Literature and Study Materials	Made available on brightspace	
Assessment	Evaluation with rubric as applied to the BEP (BSc end project)	
Special Information	The course load varies between 10 - 12 EC, depending on the 'Topic of the year' and/or the desired length of the essay.	

Year	2022/2023
Organization	Applied Sciences
Education	Master Life Science and Technology

4. Research Oriented HPM 2022

Contact for students	info-1st@tudelft.nl
Prerequisites	Master students with an excellent performance during their first semester, i.e. a GPA of 7,5 or higher. The application procedure includes a motivation letter and a short interview. After admission to the programme you will be registered for the Honours Programme Master.
Introduction 1	The Honours Programme must be approved by the Programme Director, prof.dr Ulf Hanefeld, U.Hanefeld@tudelft.nl
Program Structure 1	The student participates in a research group - different from the group where the thesis project is done, preferably outside the ChemE department or at a different university - and performs experimental and theoretical work on an actual research theme. The size and scope of the project can be chosen by mutual agreement, and varies from 1.5-2.5 months (9-15 ec). Preferably the project is combined with a course (3-6 ec) on a related topic

AS1031HPM	Applied Sciences Research Project (9-15 EC)	9
Course Coordinator	Prof.dr. J.J.C. Geerlings	
Project Coordinator	Prof.dr. J.H. van Esch	
Project Coordinator	Prof.dr. U. Hanefeld	
Project Coordinator	Dr. A.J.L. Adam	
Contact Hours / Week x/x/x/x	Different / start each period	
Education Period	1 2 3 4 Summer Holidays	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Required for	Research Oriented Honours Programme Master of the Faculty of Applied Sciences.	
Course Contents	The student participates in a research group - different from the department where the thesis project is done, preferably outside the faculty of Applied Sciences or at a different university - and performs experimental and/or theoretical work on an actual research theme. The size and scope of the project can be chosen by mutual agreement, and varies from 1.5-2.5 months (9-15 ec). Preferably the project is combined with a course (3-6 ec) on a related topic.	
Study Goals	After the project the student is able to fulfil the following requirements: - Good use of online data sources, read and critique scientific papers - Preparation and design of Experiment. Predict outcome of experiment; design experiments yourselves. - Performing experiments. Always be aware of your own and others' safety and operate accordingly; work careful and precise; correct mistakes immediately. - Analysis of data. Draw significant conclusion from the data; invite discussion of the data; evaluate the data. - Teamwork and attitude. Present at lab hours; attend your meetings; organize internal meetings and meetings with you. - Reporting. Decent labjournal is a must; clear well written report and good poster; good description of objective and conclusions.	
Education Method	Practical work, literature study, theoretical work.	
Assessment	Final mark is obtained on basis of the learning outcomes ("study goals"); see the Applied Sciences thesis grading scheme. The final mark is established by at least two scientific staff members: a) the supervisor of the research project; b) by a professor of a different department (preferably an instructor of the MSc programme of the student).	
Enrolment / Application	Master students with an excellent performance during their first semester, i.e. a GPA of 7,5 or higher, will be invited for an informative meeting in which a short overview is given of the content of the Honours Programme Master. The director of the MSc programme is responsible for admission. The application procedure includes a motivation letter and a short interview. After admission to the programme you will be registered for the Honours Programme Master.	

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